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Master Thesis - MSc. Finance and Investments

The Effect of Economic Policy Uncertainty on Cross-Border M&A Activity and Performance Perspectives of Different Types of Acquirors

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Preface

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Acknowledgements

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Abstract

This thesis examines the impact of Economic Policy Uncertainty (‘EPU’) on cross-border mergers and acquisitions (‘M&A’). Using a sample of 18 countries spread globally over the period 2004 to 2022, the effect of EPU is measured on the activity, acquisition premia, and announcement returns of international M&A deals. Furthermore, this research aims to uncover empirical evidence regarding disparities in strategic and financial acquirors on bidding behaviour and performance under elevated levels of EPU. I find sufficient evidence that higher EPU of both target and acquiror nations decrease the total number of inbound and outbound deals. I also find that target EPU enables acquirors to offer lower premia, while acquiror EPU generally increases bid premium offers. Here, I additionally find that financial acquirors reduce their deal premia more during higher EPU compared to strategic acquirors. However, the results give insufficient evidence to conclude that EPU affects deal premia. Lastly, this paper finds that on average, acquirors create larger shareholder value from deals in which EPU is high in the home domain, and lower value in the target domain, but the latter with insufficient evidence. The negative (positive) effect of EPU on announcement returns tends to be even stronger (weaker) for deals involving strategic acquirors in my sample, but similarly cannot be statistically verified. These results remain stable after performing robustness checks, including alternative variable measurements, instrumental variable analysis, and excluding highly regulated industries.

Keywords: *Economic Policy Uncertainty, Cross-Border M&A, Financial Sponsors, Strategic Acquirors, Deal Premia, Announcement Returns*

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1 - Introduction

Over the past years, major events such as the Global Financial Crisis, Brexit, the COVID-19 pandemic and the war between Russia and Ukraine have caused great political and economic instability. As a result, uncertainty in countries increased significantly, stimulating stock market volatility, and impacting future economic prospects (Al-Thaqeb & Algharabali, 2019). The uncertainty in policies that relate to economic decisions is greater than ever before, due to the globalization and interconnectedness of the world. Therefore, these events have raised the interest to understand how Economic Policy Uncertainty ('EPU') affects one of the major determinants in stimulating economic growth, the corporate investment environment. Previous research has established that higher EPU results in the delay of investments to avoid costly losses, decreasing the level of corporate investments with a depressing effect on the overall financial market (Bernanke, 1983; Bloom et al., 2007).

More specifically, a strand of literature has focused on one of the major investment decisions of a company, researching the effect of EPU on mergers and acquisitions ('M&A'). M&A is mainly driven by the need to increase value, through identifying targets with complementary technology, products, strategy, market positioning and the ability to improve efficiency (Čirjevskis, 2019; McBeath & Bacha, 2001). The greater sensitivity of takeover decisions in times of uncertainty originates from the illiquidity of these investments since M&As are generally costly and irreversible (Bonaime et al., 2018). M&A has grown to play a vital role in the world's economy, making it crucial to understand what factors impact M&A activity.

EPU not only reduces the acquisitiveness of corporations and the delay of investments but also increases the prudence with which targets are screened financially and strategically (Bonaime et al., 2018; Nguyen & Phan, 2017). Thus, prominent levels of EPU in an environment may not only reduce the quantity of M&A transactions but change the overall bidding climate and financial performance as well. Furthermore, EPU significantly impacts economic decisions including corporate dividend policy (Farooq & Ahmed, 2019), corporate risk-taking (Akey & Lewellen, 2016), and a firm's cash holdings (Phan et al., 2019).

M&A activity is dominated by the competition between financial and strategic acquirors. The two parties have fundamentally different motives to engage in M&A. These differing motives relate to time horizons, the extent of integration, strategic rationale, and their dependence on credit

conditions (Gorbenko & Malenko, 2014; Healy et al., 1997; Hege et al., 2012). However, there is limited research on what drives either party to have a more dominant position in M&A activity over time, and how their bidding behaviour and deal success are affected. Strategic acquirors seek targets with high synergy potential and long-term value creation, whereas financial sponsors place greater importance on good economic conditions and risk-spreading to generate profitable short-term returns. An uncertain economic environment comes with less favourable lending conditions, and increased complications in the valuation of a firm's future cash flows and synergy potential (Gemson et al., 2012; Lee, 2018). Therefore, the parties may have different preferences regarding targets in more uncertain environments.

Countries encountering high macroeconomic volatility tend to suffer from a weak institutional environment, leading to economic instability (Acemoglu et al., 2003; Klomp & de Haan, 2009). Governments play a primary role in macroeconomics by changing fiscal, monetary, and regulatory policies (Gregoriou et al., 2021). Pástor and Veronesi (2013) find that when the economy is weaker, governments are more likely to implement new policies. Therefore, in unstable economies, macroeconomic factors are more likely to change, and firms anticipate this by changing their levels of expenditure. Domestically, an uncertain economic environment generally slows down investments in production and employment, inducing companies to act more conservatively (Al-Thaqeb & Algharabali, 2019). However, changes in a nation's level of EPU may also incentivize firms to pursue international M&A as a means to enhance risk management, by operating in a more stable environment (Bonaime et al., 2018; Paudyal et al., 2021).

Cross-border mergers and acquisitions ('CBA') activity has rapidly increased in importance in recent decades, advancing the globalization of capital markets and driving the competitive environment internationally (Kang & Johansson, 2000). The aggregate dollar volume of acquisitions cross-border increased from 21% to 37% of total M&A activity globally from 1996 to 2014 (Cao et al., 2019). Differences between countries in terms of regulation, policies, geography, and institutional quality may stimulate or discourage cross-border takeovers and have caused merger waves globally (Ahern et al., 2015; Erel et al., 2012; Xie et al., 2017). Due to higher transaction costs and greater asymmetric information resulting from geographical distances and differences in a country's dynamics, cross-border M&As tend to be riskier than domestic takeovers (Gregoriou et al., 2021). Therefore, I expect the relationship between EPU and M&A activity to be more profound for cross-border deals. Nevertheless, how EPU influences the cross-border M&A

environment has not been exhaustively examined yet. Therefore, the research question that this study aims to answer is formulated as follows:

RQ: What is the effect of Economic Policy Uncertainty (EPU) on cross-border M&A activity and performance?

The objective of this study is to capture the acquisitive landscape of cross-border takeovers in times of high EPU. Extant literature proposes the real options theory and the hedging hypothesis, which serve as economic mechanisms to explain the relationship between EPU and international investment activity (Bernanke, 1983; Denis et al., 2002; Le & Zak, 2006). Motivated by these theories, this study employs a macroeconomic panel model, where I quantify the effect of EPU on the aggregate cross-border M&A activity of both target and acquiror nations. Furthermore, this study examines whether EPU is incorporated into investment decisions and short-term deal performance. I perform a cross-sectional analysis at the deal level to research the impact of EPU on deal premia and announcement returns. To gain a more profound understanding of the competitive M&A climate, I additionally aim to investigate the varying effects of EPU on bid premia and the performance of financial versus strategic acquirors. Hence, this study yields implications regarding which type of buyer exhibits greater sensitivity to fluctuations in a country's economic prospects.

EPU is measured according to an index constructed by Baker, Bloom and Davis (2016), hereafter referred to as the 'BBD index'. The index is newspaper-based, analysing the frequency of country-specific words related to taxes, fiscal, monetary, or regulatory policy uncertainty with economic consequences (Baker et al., 2016). Earlier studies have proven the plausibility of the BBD index to capture country-specific policy uncertainty (Al-Thaqeb & Algharabali, 2019; Bonaime et al., 2018; Nguyen & Phan, 2017; Paudyal et al., 2021).

Based on prior literature and data availability, a sample of cross-border M&A deals from 18 countries spread globally in the period from 2004 to 2022 is examined. M&A deal-level data is extracted from the Refinitiv Eikon database. Supplementary data are retrieved from Datastream, World Bank, World Factbook, CEPPI, La Porta et al. (2008) and Kenneth French (2023) databases. I find statistical evidence that a one standard deviation increase in target (acquiror) country EPU decreases the acquisition activity by 7.3% (6.3%). The transaction level analysis presents a negative reaction between target nation EPU to deal premia and announcement returns, while a positive

reaction between acquiror nation EPU on these M&A characteristics. My findings are consistent with the economic conjecture that EPU reduces a party's bargaining power to demand favourable deal conditions and increases caution about investment choices. Furthermore, I find that financial acquirors are more sensitive to EPU in acquisition premia offers, while strategic buyers facing elevated levels of EPU have more difficulty in obtaining short-term returns. However, due to the statistical insignificance of my results, there is insufficient evidence to conclude that EPU affects deal premia and shareholder value. The empirical outcomes withstand various robustness tests.

This study contributes to the previous literature focussing on factors impacting the likelihood of cross-border mergers and acquisitions, such as culture, geography, and technological advances. Furthermore, previous research has investigated the impact of EPU on stock price volatility (Brogaard & Detzel, 2015; Kundu & Paul, 2022), corporate investment decisions (Al-Thaqeb & Algharabali, 2019; Baker et al., 2016; Bloom et al., 2007) and domestic M&A (Bonaime et al., 2018; Nguyen & Phan, 2017). However, there is only limited research on the implications of uncertain environments on international acquisition decisions (Cao et al., 2019; Gregoriou et al., 2021; Paudyal et al., 2021). Therefore, this study adds to the existing research by examining the effect of EPU on cross-border mergers and acquisitions. Furthermore, to the best of my knowledge, I am the first to study discrepancies in the effect of EPU on strategic and financial acquirors relative to their bidding behaviour and short-term performance.

This study has important implications for shareholders to alter their strategies in bidding behaviour and acquisition premium offers of the target relative to their expectations of competing bidders in EPU. Furthermore, it can help both lawmakers and investors understand the effects of an uncertain environment and build overall strategies to cope with EPU. Lastly, countries that aim to attract cross-border investments could anticipate the effects by mitigating economic policy-related uncertainties.

This paper will proceed as follows. In the next section, I will discuss the theoretical framework related to prior literature to propose the hypotheses. Section 3 outlines the data, variables, and model construction. The empirical results, discussions and robustness tests are shown in Section 4. Lastly, Section 5 will give concluding remarks and discusses the limitations of this study as well as directions for future research.

2 - Literature Review

2.1 - Economic Policy Uncertainty

EPU refers to the uncertainty about who the policy decision-makers are, which fiscal, regulatory, and monetary policies will be undertaken and when, and what the economic effects of these decisions will be (Baker et al., 2016). A country's political divisions as well as the overall state of one's economy are major factors that lead to spikes in EPU (Al-Thaqeb & Algharabali, 2019). Concerns about EPU have intensified during global events such as the Global Financial Crisis (Davis, 2016), the COVID-19 Crisis (Youssef et al., 2021), and the European sovereign debt crisis (Degiannakis & Filis, 2019). However, also country-specific policy uncertainty levels have changed, through events like Brexit (Belke et al., 2018), 9/11 (Davis, 2016), or the Catalanian Independence conflict (Azqueta-Gavaldon et al., 2020). Prior research has established that governmental actions and the accompanied EPU have serious negative implications for macroeconomic outcomes (Baker et al., 2016; Bloom, 2009; McDonald & Siegel, 1986). This adverse effect can be explained both by the demand and supply side channels of credit.

On the supply side, financial markets respond to EPU through tighter credit standards, which reduces bank loans and loan growth. This in turn negatively impacts GDP levels (Bordo et al., 2016). The increasing effect of EPU on the cost of financing has two major causes. First, elevated levels of EPU lead to higher information asymmetry between firms and their creditors. This results in an increased cost of debt, as creditors want to be compensated for their informational disadvantage (Myers & Majluf, 1984; Zhang et al., 2015). Second, EPU generally leads to higher volatility in a firm's future cash flows and operating risk, which increases its default risk (Nguyen & Phan, 2017). Creditors anticipate the increased counterparty risk by raising the cost of financing (Merton, 1974; Tran, 2021).

On the demand side, higher stock market volatility in the market influences corporate behaviour, prompting businesses to reduce corporate spending, postpone investments or seek alternative investment opportunities (Al-Thaqeb & Algharabali, 2019; Tran, 2021; Zhang et al., 2015). Especially large and illiquid investments tend to be deferred (Bloom et al., 2007). One of these investments is M&A deals, which are known to be difficult and costly to reverse (Nguyen & Phan, 2017). Through unfavourable lending conditions and delays of investments, EPU may lead to steep economic downturns and slow recoveries (Baker et al., 2016; Bordo et al., 2016).

2.2 - Domestic M&A Activity

Prior academic research has explored the relationship between EPU and M&A activity. Nguyen and Phan (2017) were one of the first to empirically investigate this relationship. These authors argue that EPU impacts a firm's acquisitiveness and increases the likelihood to delay irreversible investments, resulting in a prolonged duration of M&A deal completion. Furthermore, Bonaime et al. (2018) established the negative impact of political and regulatory uncertainty on M&A activity at macro and firm levels. They argue that their findings are driven by taxes, government spending, monetary and fiscal policies, and regulation.

There are several channels through which the effect of EPU on domestic M&A activity can be explained. First, as aforementioned, uncertainty leads to an increase in the cost of debt and higher volatility in future cash flows, exacerbating firms' financial constraints (Gemson et al., 2012; Nguyen & Phan, 2017). As an increased cost of debt leads to a decrease in the acquiring shareholder value, this deters companies from raising external funds deployed to participate in M&A deals (Nguyen & Phan, 2017; Tran, 2021).

Second, the real options theory suggests that heightened uncertainty increases the value of the option to wait, resulting in the postponement of acquisitions (Bernanke, 1983; Bloom, 2009). The significance of the real options channel is conditional on the irreversibility of the deals, the costs to delay the investment, and a firm's sensitivity to policy shocks, which depends on the extent to which a firm's industry is relying on government spending and regulation (Bonaime et al., 2018).

Third, EPU forms friction in the context of M&A because the cash flow uncertainty and increased cost of debt may also enhance uncertainty over the standalone value of target companies, or the value of deal synergies (Bonaime et al., 2018). Uncertainty in macroeconomic terms increases the likelihood of a lower target's value in the interim period between deal announcement and completion, discouraging overall M&A activity (Bhagwat et al., 2016).

Lastly, an opposing perspective is that a more uncertain economic environment increases the number of deals due to reduced investor attention. EPU increases the cost of obtaining information, leading to a decrease in monitoring quality and merger analysis (Duchin & Schmidt, 2013). This limited investor attention may incentivize managers to engage in excessive corporate risk-taking and empire-building, enlarging the size, scope and influence of the organization through acquisitions (Akey & Lewellen, 2016; Bonaime et al., 2018).

2.3 - Cross-Border M&A Activity

Contrary to domestic acquisitions, cross-border takeovers are exposed to both target and acquiror country EPU, where more country-distinctive factors play a role in merger decisions. Cross-border acquirors appear to have an informational and locational disadvantage, as domestic investors have more local resources and less information asymmetry which decreases uncertainty-related costs (Paudyal et al., 2021). Higher costs impair the ability of acquirors to achieve profitable returns on their investments (Nguyen & Phan, 2017). As such, an increased EPU may thus impact cross-border M&A to a greater extent compared to domestic deals.

Previous research has focused on determinants of cross-border M&A. According to Rossi and Volpin (2004), a lower degree of investor protection in the target country incentivizes cross-border acquisitions, serving as an important channel for effective worldwide alignment in corporate governance standards. Research by La Porta et al. (2001) documents that countries with common law tend to have the strongest, and French civil law countries have the weakest shareholder protection. Furthermore, substantial research has found that geographic proximity and cultural fit affect M&A activity. The gravity model suggests that bilateral trade is positively related to GDP levels and growth, while cultural and geographical distance can increase integration costs and decrease the volume of cross-border mergers (Ahern et al., 2015; Erel et al., 2012; Hyun & Kim, 2010; Makaew, 2012). Other scholars studied the impact of macroeconomic and financial conditions as exchange rates, inflation, and trade openness concerning cross-border M&A, where weaker economies tend to be the target (Erel et al., 2012; Rossi & Volpin, 2004; Uddin & Boateng, 2011). As mentioned by Bhagwat et al. (2016), macroeconomic conditions play the most important role in affecting M&A activity.

Even though EPU in the field of cross-border M&A has not received much attention from researchers yet, recent papers from Gregoriou et al. (2021) and Paudyal et al. (2021) both investigated the impact of EPU on cross-border M&A. These authors found that when a target nation experiences high EPU, there is a decrease in the number and volume of inbound cross-border M&A. A common belief is that firms from countries with a stable political and regulatory economy are less likely to invest in countries with higher uncertainty (Holburn & Zelner, 2010). The investment deterrence view suggests that a government driven by economic nationalism may create uncertainty to deter foreign interest in domestic firms (Cao et al., 2019; Serdar Dinc & Erel, 2013). Furthermore, the delaying hypothesis implies that foreign firms time the market to avoid

uncertainties, and thus choose not to complete acquisitions before more information is available to better allocate resources (Julio & Yook, 2012).

Opposingly, opportunities may also be created for cross-border acquirors, as domestic firms become more hesitant to invest in an environment of high EPU (Al-Thaqeb & Algharabali, 2019). Thus, competition among bidders for these targets decreases. In such cases, foreign acquirors are better qualified to hedge this risk in a less uncertain environment. However, given the prevailing body of literature and earlier findings, I propose the following hypothesis:

H_{1a}: A higher level of EPU in the target nation leads to a lower number of inbound cross-border M&A deals.

Existing literature shows opposing perspectives on the relationship between high EPU in the acquiror nation and outbound cross-border M&A. On the one hand, merger decisions may be incentivized by risk management motives. In line with the hedging hypothesis, this implies that firms may be encouraged to participate in cross-border M&A as a hedging strategy by globally diversifying risk through choosing targets with lower levels of uncertainty (Cao et al., 2019; Denis et al., 2002; Le & Zak, 2006). International diversification provides acquiring companies with the opportunity to respond to differences in taxation, pricing, and other institutional differences. As such, this is considered a hedge against less beneficial implications in their home countries (Bloom et al., 2007; Cao et al., 2019).

On the other hand, a negative relationship can be inferred from the economic intuition of the real options hypothesis, in which firms become more cautious and defer especially illiquid investments when there is higher uncertainty (Bernanke, 1983; Bloom et al., 2007). In the context of cross-border M&A, there are particularly two reasons why an investment-impeding effect may occur. First, the acquirors operating in uncertain environments face higher costs of equity capital (Pástor & Veronesi, 2013). Second, targets have increased bargaining power in deal conditions to compensate for the post-merger risk of moving to a country with a high EPU (Nguyen & Phan, 2017). However, prior research established a positive relationship between high EPU of the acquiror domicile and number of outbound cross-border M&A (Gregoriou et al., 2021; Paudyal et al., 2021). Therefore, I expect these outcomes to hold within my research, and argue the following:

H_{1b}: A higher level of EPU in the acquiror nation leads to a higher number of outbound cross-border M&A deals.

2.4 - Bidding Behaviour and Performance

2.4.1 - Strategic and Financial Acquirors

Usually, takeovers tend to be competitive, in which multiple bidders negotiate with a single target (Gorbenko & Malenko, 2014). These bidders can be classified into two major but fundamentally different groups, strategic or financial acquirors. Strategic bidders, characterized by their long-term orientation, pursue targets with complementary business operations, aiming for synergistic benefits (Brogaard & Detzel, 2015; Kundu & Paul, 2022). Financial sponsors, typically pension or private equity funds, focus on short-term return and risk-spreading. These buyers seek undervalued targets with growth potential, a stable market position, and preferably a good management team (Rodgers, 2015). M&A activity of financial and strategic bidders is not synchronous, as both groups respond differently to capital market conditions, and have different time horizons and capital structures (Chiarella & Ostinelli, 2020). Therefore, I expect discrepancies between both parties' sensitivity to high EPU in terms of bidding behaviour and M&A performance.

2.4.2 - Acquisition Premium

Governmental policies shape a country's investment landscape and may affect characteristics of the M&A process, such as the acquisition premium paid (Holburn & Zelner, 2010). Deal premium describes the percentage paid over the estimated real value of a target, representing the synergies and negotiation strengths of both parties. Prior results on the relationship between EPU and deal premia have been inconclusive, whereas Nguyen and Phan (2017) found a negative relationship for domestic deals. Similarly in cross-border M&A, Gregoriou et al. (2021) found that target EPU encourages foreign acquirors to offer lower premia. In contrast, Bonaime et al. (2018) found a positive effect of EPU on deal premia.

The unpredictability surrounding government policies can complicate the valuation process of targets in the M&A process. For instance, EPU increases the volatility of a firm's future cash flows and increases default risk (Nguyen & Phan, 2017), in which it becomes more challenging to forecast the future value of a target and which increases the unpredictability of returns generated in acquisitions (Lee, 2018). Heightened levels of uncertainty surrounding a firm stimulate analysts to release more optimistic forecasts, leading to overvaluation (Ackert & Athanassakos, 1997). Furthermore, underestimating the likelihood of the adoption of unfavourable policies in the target nation could lead to overpayment in an acquisition (Lee, 2018).

As aforementioned, uncertainty leads to less favourable lending conditions, influencing acquirors to act more cautiously in the bidding process if target nations experience high uncertainty (Bernanke, 1983; Bloom, 2009; Gregoriou et al., 2021). According to Bonaime et al. (2018), this may result in higher deal premia, as the remaining acquirors are those that face significant costs if they postpone their investment, allowing targets to exert greater bargaining power to negotiate higher prices. In contrast, lower competition may weaken the bargaining position of targets, allowing foreign acquirors to demand more favourable terms (Gregoriou et al., 2021). Lee (2018) offers an alternative viewpoint, suggesting that EPU in a target nation enhances the bargaining power of foreign acquirors to demand adequate compensation for their risk exposure. Conversely, when acquirors face EPU, the target seeks compensation for the additional country risk, reducing the acquiror's bargaining power (Lee, 2018). Given these opposing views on deal premia, I examine the effect of EPU of target and acquiror nations separately, and argue the following:

H_{2a}: A higher level of EPU in the target (acquiror) nation has a negative (positive) effect on deal premium offers in cross-border M&A.

A common view is that strategic buyers generally pay higher takeover premia due to a strategic rationale and long-term benefits that may be exploited (Gorbenko & Malenko, 2014). The difference in premium is even more significant if a strategic bidder has empire-building ambitions, increasing their willingness to pay for targets (Gorbenko & Malenko, 2014). Financial acquirors prefer targets with more tangible assets, lower R&D spending and lower market-to-book ratios (Fidrmuc et al., 2012). Strategic buyers, with better access to proprietary information in negotiations and more familiarity with the target's business, can more accurately value a target than financial acquirors (Healy et al., 1997). However, most evidence suggests that higher premia are paid by strategic buyers relative to financial funds (Bargeron et al., 2008; Dittmar et al., 2012; Fidrmuc et al., 2012; Gorbenko & Malenko, 2014).

As aforementioned, EPU in the target nation could lead to complications in valuation the of the target and predictions of deal synergies (Bonaime et al., 2018). Since strategic acquirors generally benefit from valuation accuracy, EPU could weaken this economic advantage. Given that financial takeovers place less importance on synergy gains, premium offers are usually not motivated by the expectations of deal synergies (Bargeron et al., 2008). Another view suggests that a firm's bargaining strength of M&A deal terms, related to the compensation for post-merger risk in a

country with high EPU, is contingent upon the level of merger integration (Nguyen & Phan, 2017). This implies that in strategic mergers, which inherently involve complementary businesses, the implications of deal premia are more susceptible to EPU.

Other research argues that financial acquirors may be more prone to uncertainty, given that their valuations are more correlated with economic conditions (Gorbenko & Malenko, 2014). Their dependence on financial markets stems from their greater appetite for leverage and short-term focus on financial return (Tran, 2021). As the cost of financing is high during uncertainty, there is an increased difficulty to obtain profitable returns (Groh, 2009). Financial bidders are incentivized to compensate for their increased funding cost through lower premia offers, or to even withdraw their bid. Given their shorter investment horizons and greater dependence on selecting undervalued targets with a cost-reducing and growth potential (Dittmar et al., 2012), I hypothesize the following:

H_{2b}: The negative (positive) effect of EPU on the deal premium is exacerbated (moderated) by deals involving financial acquirors.

2.4.3 - Announcement Period Returns

Generally, lower bid premia lead to positive and relatively higher abnormal returns, as this increases the potential of a bidder's value creation (Hussain et al., 2022). Previous studies propose opposing views on the effect of EPU on value creation.

Some research argues that EPU increases the discretion to conduct M&A which has a higher chance of success (Nguyen & Phan, 2017). Acquiring firms tend to be more financially healthy and search for investment opportunities to strengthen their competitive position (Sha et al., 2020). In these transactions, wealth is transferred from target to bidder, in which performance increases (Paudyal et al., 2021). Furthermore, earlier research found that low political risk in a target nation increases abnormal returns for acquirors in high-risk countries (Al-Thaqeb & Algharabali, 2019).

Contrarily, other empirical evidence finds a negative relationship between EPU and performance. Shams et al. (2022) propose that EPU may incentivize acquirors to participate in empire-building ambitions, leading to lower quality deals with negative stock market reactions. Furthermore, policy uncertainty leads to higher risk and larger agency costs, impeding investment growth (Gulen & Ion, 2016; Julio & Yook, 2012). As argued before, EPU has a weakening effect on the bargaining power of a firm, as the foreign party wants to be compensated for the additional country risk (Lee, 2018). Therefore, foreign acquirors with lower EPU have increased bargaining power to demand more

favourable deal conditions, to underpay in the deal and thus capture more acquisition gains compared to targets (Gregoriou et al., 2021). However, Bonaime et al. (2018) found no statistically significant difference in returns over periods with differing levels of EPU. Given the mixed evidence on the effect of EPU on value creation in cross-border deals, I analyse target and acquiror nation EPU individually. Therefore, I argue the following:

H_{3a}: A higher level of EPU in the Target (Acquiror) nation decreases (increases) the short-term returns of acquirors in cross-border M&A deals.

As stated earlier, uncertainty adversely affects debt financing conditions, which particularly impacts highly leveraged transactions, typically undertaken by financial acquirors. Given that access to sufficient financial resources is crucial for successful post-merger integration and thus better performance (Hussain et al., 2022), strategic acquirors, who rely less on debt financing and more on internal financial resources, have a competitive advantage to obtain better performance.

On the other hand, strategic acquirors prioritise long-term value creation over immediate returns. Strategic buyers can afford to pay higher prices to close deals since they require lower rates of return and have more synergy potential (Gorbenko & Malenko, 2014). Barger et al. (2008) argue that private equity firms are more likely to walk away from a deal compared to strategic firms if the chances of failure increase. This suggests that increased EPU in deal expectations results in lower participation of financial acquirors, while strategic acquirors remain vulnerable to potentially diluted returns. Furthermore, strategic acquirors, in contrast to financial acquirors, are less dependent on fulfilling investor expectations of short-term returns (Chiarella & Ostinelli, 2020).

Shareholder gains of financial buyers depend on their ability to enhance a target's value, in connection with lucrative exit strategies and opportune market timing (Hege et al., 2009). Financial acquirors tend to use hedging strategies to spread risks and maximize firm value (Hankins, 2006). Kozol and Oran (2021) found that cross-border acquirors who employ such risk-hedging strategies obtain higher cumulative abnormal returns around deal announcements than acquirors who did not. Therefore, given all evidence, I hypothesize the following:

H_{3b}: The negative (positive) effect of EPU on value creation is exacerbated (moderated) by deals involving strategic acquirors.

In the next section, I will discuss the data collection process and describe the empirical analysis to test each of the formulated hypotheses.

3 - Methodology and Data

In this section, I will explain how the data is collected and how the relevant variables are measured. Furthermore, I report descriptive statistics of the variables of interest and present the empirical methods and models to test the formulated hypotheses.

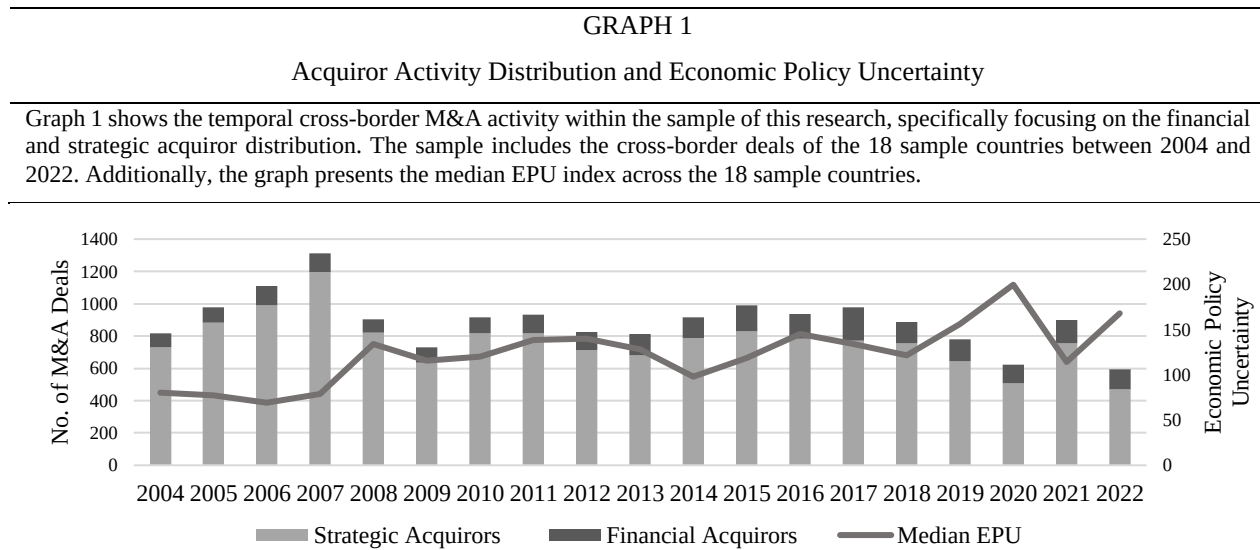
3.1 - Sample Data

The sample of M&A deals is retrieved from the Refinitiv Eikon Database. The sample includes a selection of M&A-active as well as developing nations including the following 18 countries: USA, UK, Netherlands, Germany, Japan, Canada, Australia, France, China, Spain, Hong Kong, Mexico, Russia, India, Brazil, South Korea, Colombia, and Greece. The period and country selection are based on the data availability of the national EPU indices by Baker et al. (2016). Following the methodology of Erel et al. (2012), Gregoriou et al. (2021) and Paudyal et al. (2021), the following restrictions in the dataset are applied:

- The announcement date of the deal is between January 1, 2004, and December 31, 2022;
- Target and acquiring countries belong to the 18 countries as specified before;
- Only deals with a minimum value of 1 million USD are included;
- Leverage buyouts, spin-offs, recapitalizations, exchange offers, and share repurchases will be excluded from the database;
- Only deals are included in which a majority position (above 50.01%) is acquired.

These filters yield a sample of 105,902 M&A deals. Appendix 1 presents an overview of the number of domestic and cross-border M&A deals per acquiror and target country. 15.95% of the sample accounts for cross-border M&A, including 16,896 deals. This appendix reveals that the US is the most M&A active country, with the largest number of deals both as an acquiror (36,223 deals) and as a target (37,207 deals). China and the UK are the following most-active nations in M&A. On the other hand, Colombia and Greece are the least acquisitive nations, only participating in 142 and 176 deals as an acquiror respectively. In the context of cross-border M&A, Appendix 1 demonstrates that similar to the US and the UK, both China and Canada are active in international M&A activity, both as an acquiror and target. The countries performed in 812 and 2,847 cross-border deals as acquiror, and 1,500 and 1,462 CBA deals as a target respectively. My further research is limited to only cross-border deals, where the target and acquiror domicile is in different nations.

Graph 1 illustrates the cross-border M&A activity of the constructed sample. Furthermore, the graph presents the temporal financial and strategic acquiror activity distribution and EPU median over time. Graph 1 depicts a notable association between CBA activity and EPU levels, as indicated for example in 2007 and 2020, indicating that lower EPU coincides with increased activity. Furthermore, strategic acquirors have a more prevalent presence compared to financial sponsors, and the M&A activity of financial and strategic acquirors does not show synchronous patterns.



3.2 - Variables and Measurements

3.2.1 - Independent Variable

EPU, the main variable of my study, is measured using the index created by Baker et al. (2016). The index is obtained through raw counts of country-specific words related to EPU, scaled by the total number of newspapers and standardized to unit standard deviation per month (Baker et al., 2016). The reliability of the index is checked through an extensive audit study, measuring discrepancies between human and computer-generated indices.

This index is measured over a monthly continuous horizon, and unlike political uncertainty measures, also measures policy uncertainty in non-election years (Baker et al., 2016). Political uncertainty is typically associated with events such as national elections and is measured only once every four years (Cao et al., 2019; Julio & Yook, 2012; Pástor & Veronesi, 2013). Therefore, political uncertainty as an indicator does not measure the variation in EPU between elections and does not include broader types of uncertainty related to policies (Gulen & Ion, 2016; Nguyen & Phan, 2017). The BBD index is more accurate in expressing a nation's EPU over time, rather than

based on electoral events. Additionally, prior literature has proven the impact of the index on corporate decisions such as M&A. EPU is measured for both target and acquiror nations separately, where I take the natural logarithm of the 12-month arithmetic mean of the BBD index, to create the independent variable, $EPU_{acq,tgt,t}$, of target *tgt* or acquiror *acq* country in year *t*. I lag EPU by one year to account for lags in financial decision-making by corporates as a result of uncertainty.

3.2.2 - Dependent Variables

3.2.2a - Inbound and Outbound Acquisitions

The dependent variable number of cross-border M&A is measured by taking the natural logarithm of one plus the aggregate number of cross-border deals, for target country *tgt* in year *t* denoted by $Inbound_{tgt,t}$ for Hypothesis 1a and for acquiror country *acq* in year *t* denoted by $Outbound_{acq,t}$ for Hypothesis 1b.

3.2.2b - Acquisition Premium

The main dependent variable of hypotheses 2a and 2b, M&A deal premium, denoted by $P_{i,t}$ for deal *i* at year *t* can be measured by calculating the percentage difference between the offer price and target's stock price four weeks prior to the announcement date of each deal, similar to the research of Bonaime et al. (2018) and Gregoriou et al. (2021). The interim period prevents contamination resulting from the leakage of information in the days leading up to the announcement.

3.2.2c - Announcement Returns

For the dependent variable related to Hypothesis 3a and 3b, announcement returns, I use the Event Study Tool in Datastream to calculate the market model expected returns. Since my sample consists of 18 countries worldwide, the returns of companies are related to the MSCI World Index. Following the market-adjusted model, the daily abnormal return (AR_{it}) is defined as the difference between the return of firm and the market return:

$$AR_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{Mt}) \quad (1)$$

Where AR_{it} is the abnormal return of acquiring firm *i* on day *t*, R_{it} is the actual return of acquiring firm *i* on day *t* and R_{Mt} is the market index return on day *t*. parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated by regressing the daily returns of the acquiror's stocks on the daily market index returns during the estimation window. I compute 11-day cumulative abnormal returns (CARs) of the acquiring firms over an event window $(-5, +5)$ surrounding the deal announcement date, $t = 0$.

$$CAR[-5,5] = \sum_{-5}^5 AR_{it} \quad (2)$$

To ensure accurate results in my research, I consider the possibility that companies may have run-up returns due to information leakage in the 12 trading days leading up to the announcement of a deal (Keown & Pinkerton, 1981). To account for this, I select an estimation window further away from the announcement date. Additionally, a gap between the event and estimation window limits noisy information (Castro-Iragorri, 2019). Therefore, I employ an estimation window of 60 days ending to 50 days prior to the deal announcement, to analyse how a stock covaries with the market.

3.2.3 - Financial Acquiror Flag

To compare the moderating impact of strategic and financial acquirors on deal premium and announcement returns, I use the *Financial Acquiror Flag* in Refinitiv Eikon and create a dummy variable named D_{fin} , which equals 1 if the acquiring company is buying a non-financial target company for financial reasons rather than strategic reasons.

3.2.4 - Control Variables

To account for the confounding effects of my dependent variables, I include control variables at the country-, country-pair-, deal- and firm-level in my model. An overview of the definitions, measurements, and data sources of the variables in my research are listed in Appendix 2.

Based on prior literature, I include several country control variables that may affect cross-border M&A activity. *Real GDP growth*, *GDP per capita*, *Trade Openness*, *Exchange Rate* and *Inflation* capture the annual change in a country's macroeconomic environment and economic development. Furthermore, I include *Quality of Institutions*, denoting the combined score on three dimensions including Control of Corruption, Government Effectiveness and Political Stability, which each take a value between -2.5 and 2.5. *Quality of Institutions* accounts for the degree of suboptimality in the long-term macroeconomic policies of a country. I obtain these variables from the World Bank.

I add country-pair control variables to account for bilateral characteristics influencing CBA. Two dummies for *Language* and *Religion*, retrieved from the World Factbook, control for the cultural similarity between countries. *Legal Origin* captures differences in legal protections of investors, identifying similarities in German, French or Scandinavian civil law or English common law (La Porta et al., 2008). To control for geographic proximity, I include *Geographical Distance* as a measurement for bilateral distance, and a dummy *Border* to indicate if countries are contiguous, both retrieved from the CEPII database.

Specifically for the cross-sectional analysis of hypotheses 2 and 3, I include firm and deal-characteristic variables that may have confounding effects. In line with [Gregoriou et al. \(2021\)](#), [Nguyen and Phan \(2017\)](#) and [Paudyal et al. \(2021\)](#), I include the acquiror firm-characteristic control variables *Firm Size*, *Cash Holdings*, *Leverage* and *Market-to-Book*, retrieved from Datastream. Furthermore, *Deal Size*, *Competing*, and *Tender* pertain deal level factors that could influence the acquisitiveness of an acquiror, and two dummy variables *Cash* and *Stock*, referring to the method of payment, are obtained from Refinitiv Eikon. The deal level variables are in line with the research of [Bonaime et al. \(2018\)](#), [Gregoriou et al. \(2021\)](#), and [Nguyen and Phan \(2017\)](#). Lastly, dummy *Diversifying* entails whether the target and acquiror are in other Fama-French 49 industry groups.

All control variables, except for time-invariant variables, are lagged by one year to minimize potential biases, reduce the impact of autocorrelation and ensure that the variables are exogenous to cross-border decisions. Furthermore, the presence of extreme values in my sample can drastically influence the coefficients and direction of coefficient signs ([Gregoriou et al., 2021](#); [Paudyal et al., 2021](#)). Therefore, I winsorize all data to the 1st and 99th percentile, and variables with extreme outliers to the 5th or 95th percentile. Observations with missing values are removed from my sample.

3.3 - Descriptive Statistics

Table 1 presents the summary statistics of the variables used in my research. The country-characteristics sample consists of 341 country-year observations since the EPU index of the Netherlands was not available in 2022. Furthermore, 258 country-pair observations denote the cross-border summary statistics, including only observations where a country-pair exhibits cross-border transactions. The explanatory variable, EPU, has a mean value of 4.762 and a standard deviation of 0.477. This is comparable to the findings of [Gregoriou et al. \(2021\)](#), who reported 4.710 and 0.426 as mean and standard deviation respectively. The deal level analysis includes 783 observations related to announcement returns and 672 observations for deal premia, limited to the availability of financial information of target and acquiror firms. As presented, only 4.1% of the deals involve financial acquirors. This low percentage may be attributed to the fact that acquirors and targets in this sample are publicly listed, while financial acquirors mainly include private equity funds, that acquire privately listed targets.

The premium paid on top of the target's stock price 4 weeks prior to the announcement is on average 52%. Unfortunately, since similar research by [Nguyen and Phan \(2017\)](#) and [Gregoriou et](#)

al. (2021) does not report the summary statistics of premium data, I cannot verify this value. Moreover, the eleven-day CAR appears to be close to zero, with an average return of -0.2%, but with a significant standard deviation of 13.0%.

TABLE 1
Descriptive Statistics

This table presents descriptive statistics of the key variables in the analysis of this paper. I report the mean, median, standard deviation, first and third quarters, and number of observations on country, country-pair, firm, and deal variables. The definitions of the variables are presented in Appendix 3. All variables are winsorized and time-varying variables are lagged by one year.

	Mean	Std. Dev.	Q1	Median	Q3	N
Dependent Variables						
Ln(1+CBA) _{tgt}	3.324	1.127	2.565	3.367	4.127	338
Ln(1+CBA) _{acq}	3.244	1.332	2.303	3.401	4.043	322
Deal Premium 4 Weeks Prior	52.009	56.178	23.270	39.005	66.580	672
CAR Market Model Acq [-5,+5]	-0.002	0.130	-0.064	0.002	0.064	783
Country Characteristics						
Ln(EPU)	4.762	0.477	4.416	4.733	4.127	341
GDP Growth	0.025	0.035	0.118	0.024	0.041	341
Ln(GDP/Capita)	9.903	0.965	9.121	10.354	10.656	341
Trade Openness	0.743	0.748	0.395	0.561	0.698	341
Inflation	0.028	0.022	0.138	0.024	0.038	341
Ln(Exchange Rate)	1.840	2.444	-0.133	0.540	3.458	341
Quality of Institutions	1.900	2.471	-0.885	2.955	4.138	341
Cross-Border Characteristics						
Ln(Geographic Distance)	8.670	0.936	8.376	9.000	9.195	258
Border (0-1)	0.070	0.255	0	0	0	258
Legal Origin (0-1)	0.318	0.467	0	0	1	258
Language (0-1)	0.078	0.268	0	0	0	258
Religion (0-1)	0.136	0.343	0	0	0	258
Acquirer-Firm Characteristics						
Firm Size [ln(Assets)]	14.699	2.501	13.109	15.068	16.640	783
Leverage	0.202	0.159	0.054	0.182	0.312	783
Cash Holdings	0.174	0.186	0.060	0.119	0.233	783
Market-to-Book	2.689	2.490	1.390	2.200	3.500	783
Deal Characteristics						
Financial Acquiror (0-1)	0.041	0.198	0	0	0	783
Ln(Deal Value)	5.723	2.107	4.139	5.732	7.250	783
Diversifying Deal (0-1)	0.409	0.492	0	0	1	783
Cash Payment (0-1)	0.575	0.495	0	1	1	783
Stock Payment (0-1)	0.179	0.383	0	0	0	783
Competing Offer (0-1)	0.070	0.256	0	0	0	783
Tender Offer (0-1)	0.282	0.450	0	0	1	783

3.4 - Model Construction

3.4.1 - Cross-Border Deals

To answer the first hypothesis, the effect of EPU on the number of cross-border M&A deals is analysed. I investigate whether the CBA of the target country (inbound) and acquiror country (outbound) are influenced by changes in EPU. Following Cao et al. (2019) and Ferreira et al. (2010), a panel regression analysis is performed to test the relationship between a country's EPU

on both the deal inflow and outflow of cross-border M&A. The regressions performed to test H_{1a} and H_{1b} is:

$$H_{1a} \quad \sum_{i=1}^N Inbound_{tgt,t} = \beta_0 + \beta_1 EPU_{tgt,t-1} + \beta_2 X_{tgt,t-1} + \alpha_i + \lambda_t + \varepsilon_{i,t} \quad (3)$$

$$H_{1b} \quad \sum_{i=1}^N Outbound_{acq,t} = \beta_0 + \beta_1 EPU_{acq,t-1} + \beta_2 X_{acq,t-1} + \alpha_i + \lambda_t + \varepsilon_{i,t} \quad (4)$$

The main explanatory variable, EPU is measured for both target and acquiror country in year $t - 1$ and denoted by $EPU_{tgt,t-1}$ and $EPU_{acq,t-1}$ respectively. X denotes the vector of country and country-pair control variables, as presented in the summary statistics of Table 1. All specifications include α_i country fixed effects and λ_t year fixed effects.

For hypotheses 2 and 3, a cross-sectional regression is performed using data from Refinitiv Eikon with similar restrictions as mentioned in section 3.1. Here, I differentiate between the types of acquirors and examine their moderating effect on deal premium and announcement returns.

3.4.2 - Deal Premium

To obtain financial information on acquisition premia for Hypothesis 2, I have imposed an additional criterion for this new sample, requiring targets to be publicly listed. A cross-sectional regression analysis is performed to research the relationship between changes in EPU either the acquiror or target country on the deal premium $P_{i,t}$ of deal i in year t . X includes a vector of country, country-pair, firm, and deal-level control variables. In the following specifications, next to α_i country fixed effects and λ_t year fixed effects, σ_d industry-fixed effects are included. Thus, the regression for H_{2a} is structured the following:

$$H_{2a} \quad P_{i,t} = \beta_0 + \beta_1 EPU_{acq,tgt,t-1} + \beta_2 X_{i,t-1} + \alpha_i + \lambda_t + \sigma_d + \varepsilon_{i,t} \quad (5)$$

For H_{2b} , I test whether financial acquirors pay lower premia during higher levels of EPU of target and acquiror nations, presented by the following regression:

$$H_{2b} \quad P_{i,t} = \beta_0 + \beta_1 EPU_{tgt,acq,t-1} + \beta_2 D_{fin,t} + \beta_3 (EPU_{tgt,acq,t-1} * D_{fin,t}) + \beta_4 X_{i,t-1} + \alpha_i + \lambda_t + \sigma_d + \varepsilon_{i,t} \quad (6)$$

A dummy variable, D_{fin} , which equals 1 for financial acquirors and 0 for strategic acquirors, is included to measure this effect. The interacting effect of the type of acquiror and EPU, captured by the interaction term $EPU_{tgt,acq,t-1} * D_{fin,t}$, is also included to allow for a changing level of impact of EPU of financial and strategic acquirors on deal premia.

3.4.3 - Announcement Returns

To measure the effect of EPU on the announcement returns of cross-border acquirors, I have imposed an additional sample restriction, requiring both targets and acquirors to be publicly listed. A cross-sectional regression is performed to measure the relationship between EPU and the acquiror's CARs. X denotes the vector of all country, country-pair, firm, and deal-level control variables. The regression for H_{3a} is:

$$H_{3a} \quad CAR[-5,5] = \beta_0 + \beta_1 EPU_{acq,tgt,t-1} + \beta_2 X_{i,t-1} + \alpha_i + \lambda_t + \sigma_d + \varepsilon_{i,t} \quad (7)$$

To measure whether EPU more negatively affects strategic acquirors in creating short-term value, the following regression to test H_{3b} is performed:

$$H_{3b} \quad CAR[-5,5] = \beta_0 + \beta_1 EPU_{acq,tgt,t-1} + \beta_2 D_{fin,t} + \beta_3 (EPU_{acq,tgt,t-1} * D_{fin,t}) + \beta_4 X_{i,t-1} + \alpha_i + \lambda_t + \sigma_d + \varepsilon_{i,t} \quad (8)$$

The dummy variable D_{fin} , which equals 1 for strategic buyers and 0 for financial buyers is included in the regression, as well as its interaction effect with EPU, captured by $EPU_{acq,tgt,t-1} * D_{fin,t}$.

3.4.4 - Reliability of Results

To correctly specify the models used in my analysis, I run several tests to address potential multicollinearity, heteroskedasticity, autocorrelation and fixed-effects concerns. Furthermore, I introduce the implications of the 2SLS estimation and regulated industries robustness tests.

The first test investigates multicollinearity, by computing a pairwise correlation matrix on all explanatory variables and performing a VIF test. The VIF test results in individual scores below 10 each, and I detect no substantial correction (>0.8) in the correlation matrix. Thus, I conclude no highly correlated explanatory variables.

To test potential heteroskedasticity, I perform the Modified Wald test for groupwise heteroskedasticity, related to the panel regressions in Hypothesis 1, and the Breusch-pagan test on my cross-sectional models in Hypothesis 2 and 3. Since the p -value of these tests is consistently significant (<0.05), the null hypothesis of homoskedasticity can be rejected and robust standard errors need to be included in each of my models.

Furthermore, the panel regression models related to H_{1a} and H_{1b} are tested for autocorrelation with the Breusch-Godfrey/Wooldridge test. The null hypothesis, suggesting no serial correlation, is

violated, meaning that the standard errors are misleading and need to be corrected. The cross-sectional regressions in the analysis of deal premium and announcement returns are not tested on autocorrelation since each observation represents an independent event.

To capture the effect of unobserved variables in the error term that influence M&A activity, I perform the Wald test on the year- country- and industry level. I can reject the null hypothesis suggesting no significant impact of fixed effects ($p < 0.05$). Therefore, I will include year and country-fixed effects for the panel regressions, similar to the research of Gregoriou et al. (2021) and Paudyal et al. (2021) to mitigate disparities in country-specific and macroeconomic factors. For the cross-sectional regressions, I additionally include industry-fixed effects SIC at the two-digit level, following Cao et al. (2019).

To investigate whether EPU may be an endogenous variable, I will employ an instrumental variable (IV) model through a two-least squares estimation (2SLS), enabling me to address omitted variables and reverse causality issues. The instrumental variable (IV) needs to be correlated with EPU but uncorrelated with the error term, thus I use the *Regulatory Quality* indicator (retrieved from the World Bank), describing a government's ability to formulate and implement sound policies and regulations. Baker et al. (2016) document that regulatory uncertainty is one of the main drivers of EPU and thus presents a conceptually negative relationship. Furthermore, I argue that a government's capacity to enforce effective policies is not directly linked to CBA and is thus exogenous from the error term. The results of the analysis are presented in [Section 4.4.2](#)

While the effect of EPU has widespread implications for various industries in the economy, sectors heavily regulated by the government tend to be more exposed to significant policy risks (García-Canal & Guillén, 2008). Regulated industries experience an exceptional level of intervention, where governments may drastically alter investment strategies and the profitability of companies (Henisz & Zelner, 2001). To mitigate potential biases introduced by regulated industries that may dominate the relationships found in previous results, I investigate the effects of EPU on deal premium and eleven-day CARs using a subsample excluding the most heavily regulated industries; utility (SIC codes 4900-4999) and financial industries (6000-6999), following Nguyen and Phan (2017). The outcomes of the subsample analysis are provided in [Section 4.4.4](#).

4 - Results and Discussions

In this section, the results will be discussed following the three hypotheses. I will first present the impact of EPU on the number of inbound and outbound cross-border M&A deals, followed by the analysis of the impact of EPU on deal premium and short-term value creation, where I also present the interaction effect of different types of acquirors. Finally, I perform several robustness tests.

4.1 - EPU and Number of Cross-Border Acquisitions

Table 2 presents the panel regressions on target country (inbound) and acquiror country (outbound) acquisitions. The panel regressions are unbalanced, given that not every country in the sample participated in cross-border M&A deals in each year from 2004 to 2022 within my sample.

Columns (1) - (3) of Table 2 present the results on the inbound cross-border M&A activity related to H_{1a} , expecting a negative relationship related to the delaying hypothesis. Column (1) measures the number of M&A deals with EPU as single independent variable, and columns (2) and (3) include additional control variables. Each model presents a negative and significant relationship between EPU in the target nation and the number of inbound deals, following H_{1a} . After including control variables and fixed effects, the coefficient of EPU is negative with a value of 0.152, significant at a 1% level. Related to the economic significance, a one standard deviation increase in EPU (0.477) is associated with a decrease in the inbound deal number of 7.3% (0.152×0.477).

Similarly, columns (4) – (6) present the impact of EPU on outbound cross-border M&A, related to H_{1b} , expecting a positive relationship given the hedging motives of acquirors with higher EPU. A negative and significant (10%) relationship is found between EPU in the acquiror nation and the number of outbound deals. Column (6), demonstrating the analysis including all control variables, year- and country-fixed effects present a coefficient of -0.132. Economically, this relationship shows that a one standard deviation increase in EPU leads to a decrease in outbound deals of 6.3% (0.132×0.477). Furthermore, columns (5) and (6) reveal evidence of the positive relationship between GDP per capita, GDP growth and Trade Openness and the number of cross-border deals at a 1% significance level, in line with prior literature of Hyun and Kim (2010), and Uddin and Boateng (2011). In addition, a significant and negative relationship (at the 1% level) is observed between the Exchange Rate and outbound deals. This finding may suggest that higher exchange rates in the acquiror nation increase the cost of cross-border acquisitions, making acquisitions less attractive and thus leading to a lower number of M&A deals.

TABLE 2
EPU and Cross-Border Deal Analysis

Table 2 presents the effect of Economic Policy Uncertainty on cross-border M&A activity of 18 countries between 2004 and 2022. The dependent variable in columns (1) – (3) presents the natural logarithm of one plus the number of CBA of the target nation. The dependent variable in columns (4) – (6) presents the natural logarithm of one plus the number of CBA in the acquiror nation. Robust standard errors in parentheses to adjust for autocorrelation and heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.

VARIABLES	Inbound Cross-Border M&A Deals			Outbound Cross-Border M&A Deals		
	(1)	(2)	(3)	(4)	(5)	(6)
Ln(EPU) _{tgt}	-0.164**	-0.136**	-0.152***			
	-0.065	(0.055)	(0.055)			
Ln(EPU) _{acq}				-0.089	-0.136*	-0.132*
				-0.085	(0.075)	(0.078)
Country Characteristics						
GDP growth		-1.592*	-1.458		3.706***	3.738***
		(0.930)	(0.930)		(1.274)	(1.286)
Ln(GDP/capita)		-0.173	-0.164		0.958***	0.893***
		(0.222)	(0.223)		(0.308)	(0.306)
Trade Openness		0.825***	0.784***		0.647***	0.639***
		(0.247)	(0.229)		(0.210)	(0.214)
Inflation		0.721	0.738		2.954	2.954
		(1.820)	(1.815)		(2.444)	(2.466)
Ln(Exchange Rate)		-0.913***	-0.891***		-0.791***	-0.798***
		(0.160)	(0.157)		(0.248)	(0.249)
Quality of Institutions		0.043	0.046		-0.007	-0.005
		(0.057)	(0.056)		(0.076)	(0.078)
Country-Pair Characteristics						
Ln(Geographic Distance)			-0.003			-0.010
			(0.022)			(0.022)
Border (0 - 1)			-0.011			-0.071
			(0.072)			(0.074)
Legal Origin (0 - 1)			-0.098**			0.036
			(0.048)			(0.057)
Language (0 - 1)			0.095			-0.080
			(0.060)			(0.075)
Religion (0 - 1)			0.065			0.010
			(0.075)			(0.069)
Constant	4.107***	6.682***	6.684***	3.665***	-4.850	-4.112
	-0.308	(2.258)	(2.290)	-0.405	(3.104)	(3.091)
Country FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	338	338	338	322	322	322
Adjusted R-squared	0.922	0.933	0.933	0.916	0.932	0.932

In conclusion, the results in Table 3 are in line with the expectations on H_{1a} , suggesting that targets with high levels of EPU tend to attract less cross-border M&A. This relationship is supported by prior literature of [Cao et al. \(2019\)](#), [Gregoriou et al. \(2021\)](#), [Holburn and Zelner \(2010\)](#) and [Paudyal et al. \(2021\)](#), corroborating this relationship through the delaying hypothesis and the deterrence view. Contrary to expectations on H_{1b} , implying that higher EPU in acquiror nations leads to increased engagement in outbound deals, my results present a statistically significant negative relationship. This is against the literature of [Cao et al. \(2019\)](#) and [Gregoriou et al. \(2021\)](#), suggesting that acquirors seek hedging mechanisms as a risk-reducing strategy through acquiring

a cross-border target. The negative relationship may imply that the hedging strategy does not play a dominant role within my sample. As an alternative explanation, the real options hypothesis also supports the negative effect (Bernanke (1983); Bloom et al. (2007)).

4.2 - EPU and Deal Premium

In this section, I will examine the impact of EPU on deal premia relative to the target's stock price 4-weeks prior to the announcement.

Columns (1) and (2) present the effect of EPU on deal premia. In economic terms, a one standard deviation increase in EPU (0.477) of the target country leads to a 1.15% decline in premium offer, after including all control variables. Columns (3) and (4) show the relationship between acquiror nation EPU and deal premium offers. After including control variables, the magnitude of the effect decreases, as the coefficient of EPU falls from 4.772 to 0.367, but remains positive. The outcome suggests that a one standard deviation increase in EPU in the acquiror nation leads to a 0.18% increase in premium offers. However, these relationships are statistically insignificant.

Moreover, positive, and significant relationships are found between the market-to-book ratio of the acquiror (at 5% level), and competing offers (at 1% level), describing that high market-valued acquirors, as well as a competitive deal environment, drive up deal premia. These relationships were also found by Gregoriou et al. (2021), and Nguyen and Phan (2017). The analysis further reveals a negative relationship for the Financial Acquiror Flag, aligning with literature suggesting that financial acquirors, who do not factor synergy considerations, typically offer lower premia.

Table 4 presents the analysis of Hypothesis 2b, to investigate whether financial acquirors are more sensitive in adjusting their bidding premiums in response to EPU than strategic acquirors. I include interaction term $\ln(EPU) * Financial\ Acquiror$ to allow for changes in moderating impact of financial acquirors in the relationship between EPU and deal premia.

After incorporating the interaction term, the direction of the relationships between EPU and deal premia remain consistent. Column (1) and (2) again present a negative influence of EPU of targets on deal premia, and displays negative, and significant (at 10% level) coefficients of the interaction term, amounting to -25.817 and -23.816. After including control variables, economically, this implies that a one standard deviation increase in target EPU generally decreases the deal premia of financial acquirors by 14.81% ($0.477 * (-7.241 - 23.816)$), while strategic acquirors only decrease the

deal premia by 6.76% (0.477*-7.241). Columns (3) and (4) similarly report a positive relationship between acquiror EPU and bid premia and present negative but insignificant coefficients for the interaction term. Economically, after including control variables, a one standard deviation increase in acquiror EPU decreases the deal premium offers of financial acquirors by 4.87% (0.477*(0.632-10.732), while strategic acquirors increase their premium offer by 0.3% (0.477*(0.632).

TABLE 3
EPU and Deal Premia

The table presents the effect of Economic Policy Uncertainty on the deal premia of 18 countries between 2004 and 2022. The dependent variable is defined as the percentage difference between the offer price over the target's stock price 4 weeks prior to the deal announcement. EPU is measured for the target country in columns (1) and (2), with target country control variables, and measured for the acquiror country in columns (3) and (4), using acquiror country control variables. Country, year, and industry fixed effects are included. Robust standard errors in parentheses to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.

VARIABLES	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-1.242 (5.862)	-1.630 (6.209)		
Ln(EPU) _{acq}			4.772 (9.115)	0.367 (9.112)
Acquiror Firm Characteristics				
Firm Size		-0.960 (2.516)		-1.456 (1.649)
Leverage		7.875 (25.237)		16.954 (21.520)
Cash		17.830 (11.464)		15.245 (17.058)
Market-to-Book		1.529** (0.656)		1.953 (1.547)
Deal Characteristics				
Ln(Deal Value)		-6.022** (2.701)		-5.711*** (1.729)
Financial Acquiror (0 – 1)		-17.927** (7.404)		-26.742*** (9.915)
Cash Payment (0 – 1)		2.623 (4.781)		(4.945)
Stock Payment (0 – 1)		-1.902 (7.858)		-0.671 (9.328)
Competing Offer (0 – 1)		28.103*** (5.461)		28.949*** (10.075)
Tender Offer (0 – 1)		6.082 (4.330)		6.385 (5.228)
Diversifying Deal (0 – 1)		3.738 (5.190)		4.993 (5.154)
Constant	58.016* (28.096)	-17.459 (50.762)	28.965 (44.238)	264.649** (133.517)
Country(-pair) Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	672	672	672	672
Adjusted R-squared	0.023	0.079	0.004	0.072

TABLE 4				
EPU and Deal Premium – Interaction Financial Acquiror				
This table presents the interacting effect of the Financial Acquiror Flag and Economic Policy Uncertainty on 4-week premium offers in cross-border M&A deals in 18 sample countries during the period 2004 to 2022. Columns (2) and (4) also include control variables. EPU is measured for the target country in columns (1) and (2), with target country control variables, and measured for the acquiror country in columns (3) and (4), using acquiror control variables. Country, year, and industry fixed effects are included. Robust standard errors in parentheses to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.				
VARIABLES	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-15.273 (9.729)	-7.241 (9.935)		
Ln(EPU) _{acq}			4.688 (8.968)	0.632 (9.143)
Ln(EPU) _{tgt} * Financial Acquiror	-25.817* (15.353)	-23.816* (12.079)		
Ln(EPU) _{acq} * Financial Acquiror			-12.094 (17.280)	-10.732 (16.643)
Financial Acquiror (0 – 1)	109.745 (74.283)	95.772 (61.990)	35.879 (72.253)	26.536 (84.835)
Constant	126.940*** (47.707)	-120.354 (222.818)	30.690 (43.529)	266.536*** (133.569)
Control variables	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	672	672	672	672
Adjusted R-squared	0.023	0.060	0.019	0.071

In conclusion, results in Tables 3 and 4 both report a negative (positive) relationship between the target (acquiror) EPU and premium offers, supporting Hypothesis 2a. The negative relationship is supported by prior literature of Gregoriou et al. (2021), Lee (2018), and Nguyen and Phan (2017), proposing an increased bargaining power of the acquiror as well as more caution in bidding prices when targets have higher EPU. The positive relationship between acquiror EPU and deal premia was also found in the literature of Bonaime et al. (2018) and Lee (2018), relating to the increased bargaining power of the target to compensate for the uncertainties of the acquiror.

Furthermore, in line with Hypothesis 2b, I find that financial acquirors decrease their offer premia more significantly than strategic bidders when targets experience higher EPU (at a 10% significance level), which may suggest that financial acquirors are more cautious in bidding behaviour given their greater dependence on economic conditions, and thus their anticipation on the increased difficulty in obtaining profitable returns (Gemson et al., 2012; Groh, 2009).

However, given the insignificance of the EPU coefficients, my data provides insufficient evidence to reject the null hypothesis that EPU has no impact on deal premia offers in cross-border deals. This may be caused by a small sample size resulting in low variability in data. Given that the sample size is substantially reduced in my analysis due to the availability of the BBD index and deal premia

and financial information availability, this explanation may be plausible to explain the insignificance. Table 4 also shows notable extreme coefficients for the interaction term and financial acquiror dummy. The limited representation of financial acquirors in the sample could bias the estimates and may inflate the value of these coefficients. Furthermore, heterogeneity among financial acquirors, related to strategies, risk profiles and characteristics may introduce increased variability of results and thus result in larger coefficients in the regression analysis.

4.3 - EPU and Acquiror Announcement Returns

For Hypothesis 3a, I investigate the effect of EPU on the market model eleven-day CAR earned by acquirors. The results are presented in Table 5.

Following expectations, a negative relationship between EPU in the target nation and acquiror announcement returns is found in columns (1) and (2), where the coefficients amount to -0.020 and -0.011 respectively. Economically, after including all control variables, a one standard deviation increase in the EPU of the target nation leads to a 0.52% (0.477×-0.011) decrease in acquiror abnormal returns. However, no valid conclusions can be drawn since the results are statistically insignificant. Columns (3) and (4) on the other hand reveal positive and statistically significant results (at a 10% and 5% level respectively), demonstrating that increased EPU in the acquiror nation results in a positive stock market reaction. After including control variables (column 4), the magnitude of the EPU coefficient increases from 0.041 to 0.054, with the economic implication that a one standard deviation increase in acquiror EPU results in a 2.6% (0.477×0.054) increase in acquiror returns.

Next, I examine the interacting effect of different types of acquirors related to Hypothesis 3b, where I expect a stronger (weaker) negative (positive) effect of target EPU on strategic acquirors' announcement returns. Table 6 includes the interacting effect of financial acquirers and EPU using the interaction term $\text{Ln}(EPU) * \text{Financial Acquiror}$ on the acquiror's eleven-day CARs.

Consistent with the results of Table 5, with control variables, the relationships between CARs and target EPU remain negative and insignificant, and for acquiror EPU positive and significant (at the 5% level). The interaction term presents a positive coefficient for both columns (1) and (2), where the interaction with a target EPU is significant (at a 5% and 10% level). Economically for column (2), a one standard deviation increase in EPU increases the CAR of financial acquirors by 3.8%

(0.477*(-0.014+0.094), while decreases the CAR of strategic acquirors by 0.7% (0.477*-0.014). Related to acquiror EPU in columns (3) and (4), I also find a positive but insignificant relationship between the interaction term and deal premia. The coefficients of the interaction term amount to 0.057 and 0.044 respectively, where column (4) economically shows that a one standard deviation increase in EPU increases the CAR of financial acquirors by 7.4% (0.477*(0.053+0.044)) while of strategic acquirors only by 2.5% (0.477*0.053). The findings are in line with Hypothesis 3b, but the statistical insignificance leads to insufficient evidence to reject the null hypothesis that there is no difference in the effect of EPU on strategic and financial acquiror CARs.

TABLE 5
EPU and Announcement Returns

This Table presents the effect of Economic Policy Uncertainty on the announcement returns of the acquiror of 18 countries from 2004 to 2022. The dependent variable is the cumulative abnormal return (CAR) on an eleven-day event window. The main explanatory variable is measured for the target country in columns (1) and (2), with target country-fixed effects, and measured for the acquiror country in columns (3) and (4), using acquiror country-fixed effects. The results include year-, country-, and industry-fixed effects. Robust standard errors are included to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.

VARIABLES	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-0.020 (0.025)	-0.011 (0.026)		
Ln(EPU) _{acq}			0.041* (0.024)	0.054** (0.024)
Acquiror Firm Characteristics				
Firm Size		-0.000 (0.004)		0.002 (0.004)
Leverage		0.020 (0.039)		-0.001 (0.042)
Cash		0.027 (0.042)		0.037 (0.042)
Market-to-Book		0.003 (0.003)		0.004 (0.003)
Deal Characteristics				
Ln(Deal Value)		-0.002 (0.004)		-0.005 (0.004)
Financial Acquiror (0 – 1)		0.040 (0.026)		0.049* (0.027)
Cash Payment (0 – 1)		0.012 (0.014)		0.012 (0.013)
Stock Payment (0 – 1)		-0.019 (0.020)		-0.012 (0.021)
Competing Offer (0 – 1)		-0.009 (0.019)		-0.013 (0.018)
Tender Offer (0 – 1)		-0.007 (0.012)		-0.008 (0.011)
Diversifying Deal (0 – 1)		0.033*** (0.012)		0.032*** (0.011)
Constant	0.097 (0.121)	1.722 (3.815)	-0.200* (0.117)	1.149 (2.133)
Country(-pair) Control Variables	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	783	783	783	783
Adjusted R-squared	-0.020	-0.010	-0.012	-0.003

TABLE 6				
EPU and Announcement Returns - Interaction Financial Acquiror				
This table presents the interacting effect of the Financial Acquiror Flag and Economic Policy Uncertainty on the shareholder value of acquirors of cross-border M&A deals of 18 sample countries from 2004 to 2022. Columns (2) and (4) also include control variables. EPU is measured for the target country in columns (1) and (2), with target country control variables, and measured for the acquiror country in columns (3) and (4), with acquiror control variables. Country, year, and industry fixed effects are included, and robust standard errors in parentheses to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.				
	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-0.022 (0.025)	-0.014 (0.026)		
Ln(EPU) _{acq}			0.038 (0.025)	0.053** (0.025)
Ln(EPU) _{tgt} * Financial Acquiror	0.100** (0.047)	0.094* (0.052)		
Ln(EPU) _{acq} * Financial Acquiror			0.057 (0.039)	0.044 (0.043)
Financial Acquiror (0 – 1)	-0.454** (0.221)	-0.422* (0.244)	-0.239 (0.193)	-0.174 (0.216)
Constant	0.101 (0.121)	1.432 (3.879)	-0.189 (0.119)	0.510 (2.133)
Control variables	No	Yes	No	Yes
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Observations	783	783	783	783
Adjusted R-squared	-0.015	-0.018	-0.010	-0.015

In conclusion, I find a negative and insignificant stock market reaction to M&A deals that include a target with higher EPU, in line with Hypothesis 3a. This relationship is supported by Gregoriou et al. (2021), Gulen and Ion (2016), and Julio and Yook (2012), who suggest that higher risk increases principal-agent costs and complicates firm valuations, hampering investment growth and value creation. This contradicts the proposition of Lee (2018), implying that acquirors have higher bargaining power to capture a larger share of the gains. Furthermore, I find a positive and significant stock market reaction to M&A deals with high acquiror EPU, in line with findings of Nguyen & Phan (2017) and Sha et al. (2020), who suggest that EPU leads to increased prudence to conduct M&A with better expected outcomes. This is contrary to the finding of Shams et al. (2022), proposing that acquirors may be incentivized to participate in empire-building ambitions, resulting in lower returns. Consistent with Hypothesis 3b, I find that financial sponsors generally obtain higher announcement returns than strategic acquirors if the acquiror or target experiences EPU. This corroborates with the literature of Chiarella and Ostinelli (2020), and Gorbenko and Malenko (2014), who explain that strategic acquirors are less dependent on short-term gains, and can afford lower returns given future synergy potential. However, most results provide insufficient proof to reject the null hypothesis that EPU does not affect shareholder announcement returns. Again, I argue that insignificance may originate from low data variability caused by a small sample size.

4.4 - Robustness Tests

In this section, several robustness checks are performed to verify the consistency of prior results. First, I reflect on high and low levels of EPU related to the EPU median of cross-border M&A activity. Second, I perform a two-least squares analysis to account for potential endogeneity in the model. Third, alternative measures for the dependent variables Deal Premium and CARs are examined. Lastly, I perform a sub-analysis to consider potential biased estimates caused by regulated industries.

4.4.1 - EPU High-Low Matrix

I apply alternative methods to proxy for EPU as a robustness test. I use the *EPU Median* as explanatory variable in the first method, to account for the possibility that EPU is correlated across countries. I do not include year-fixed effects, given that both account for cross-sectional correlation over time and are thus collinear. Furthermore, I create dummy variables $EPU_high_{i,t}$, which equals 1 if the EPU of country i at year t is higher than EPU_M_t , and $EPU_low_{i,t}$ in case of a lower EPU level. Using combinations of the varying levels of EPU for both targets and acquirors, I create four distinct matrix options, as presented in Table 7.

TABLE 7 EPU High-Low Matrix								
This table presents the robustness tests of the Economic Policy Uncertainty variable, using variation in high and low levels of EPU related to the EPU Median. The dependent variable of Panel A (B) is the natural logarithm of one plus the number of inbound (outbound) M&A deals. Columns (1) and (5) present the effect of the EPU median on CBA. Columns (2)-(4) and (5)-(8) present the variation in the high and low levels of EPU. Country- and year-fixed effects are included. The robust standard errors are included to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.								
VARIABLES	Panel A: Inbound M&A Activity				Panel B: Outbound M&A Activity			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
EPU_M	-0.433*** (0.097)				-0.346*** (0.126)			
EPU _{high, high}		0.012 (0.047)				-0.091* (0.050)		
EPU _{high(tgt), low(acq)}			-0.086* (0.051)					
EPU _{low(tgt), high(acq)}							-0.054 (0.054)	
EPU _{low, low}				-0.061 (0.060)				0.186*** (0.070)
Constant	7.144*** (1.859)	2.942 (2.338)	2.997 (2.351)	3.067 (2.315)	1.173 (2.833)	-4.625 (3.173)	-4.492 (3.220)	-4.984 (3.224)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Observations	338	338	338	338	322	322	322	322
Adj. R-squared	0.912	0.921	0.922	0.921	0.913	0.929	0.928	0.930

The results in columns (1) and (5) reconfirm the results in the previous findings on the relationship between EPU and inbound and outbound CBA, demonstrating a negative and significant relationship at 1% significance. Furthermore, column (3) shows that for a target in high EPU, with an acquiror in low EPU, the negative effect pertains to the volume of inbound acquisitions, supporting the earlier findings. Interestingly, for outbound M&A activity, significant results are found only for the high-high (6) and low-low (8) combinations. If both the EPU of target and acquiror country is high, the acquisitiveness of firms tends to decrease at a 10% significance level, while if both countries have a below-median EPU level, the number of outbound deals increases at 1% significance.

4.4.2 - Two-Least Squares Estimation

To address the endogeneity concerns related to EPU on cross-border M&A, I perform the two-stage least squares (2SLS) estimation, using *Regulatory Quality* as instrumental variable. The results are presented in Table 8. The first-stage regression in column (1) gives evidence of the quality of the IV, exhibiting a negative and statistically significant relationship (at a 1% level), as was theoretically expected. In the second-stage regression in columns (2) and (3), I perform similar regressions to columns (3) and (6) of Table 2, but now using the obtained fitted values of EPU, uncorrelated with the error term ε to obtain unbiased estimates. I find that for inbound acquisitions, the impact of the fitted values of EPU remains negative and significant, confirming previous

TABLE 8
Two-Stage Least Squares Estimation

This table presents a two-stage least squares (2SLS) estimation to address endogeneity issues related to Economic Policy Uncertainty. In column (1) I present the first-stage regression estimates in a two-stage least squares (2SLS) estimation. The dependent variable in column (1) is EPU, and Regulatory Quality is used as the IV. The dependent variable in specifications (2) and (3) are the natural log of one plus the total number of cross-border M&A deals. Year- and country-fixed effects are included. Robust standard errors are included to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.

VARIABLES	Inbound cross-border M&A deals		Outbound cross-border M&A deals	
	2SLS 1st stage (1)	2SLS 2nd stage (2)	2SLS 2nd stage (3)	
EPU_fitted		-0.673** (0.261)	-0.249 (0.314)	
Regulatory Quality	-0.583*** (0.124)			
Constant	1.460 (2.005)	4.486* (2.345)	-3.924 (3.313)	
Control Variables	Yes	Yes	Yes	
Country FE	Yes	Yes	Yes	
Year FE	Yes	Yes	Yes	
Observations	338	338	322	
Adjusted R-squared	0.655	0.923	0.931	

findings. For outbound acquisitions, still, a negative but now insignificant relationship is presented. Therefore, through using the instrumental variable analysis in the 2SLS estimation, endogeneity is adequately addressed. The results remain stable, where the endogeneity bias is reduced through more reliable and unbiased estimates for the causal relationship between EPU and cross-border M&A activity.

4.4.3 - Alternative Dependent Variable Measurements

Appendix 3 shows alternative measures of deal premium related to hypothesis 2, where columns (1) and (2) measure the percentage overpayment of the target's stock price one week prior and columns (3) and (4) one day prior to the deal announcement. These measures yield comparable results to those in Table 3, presenting a positive (negative) relationship between the target (acquiror) EPU and deal premium but again lacking statistical significance.

The dependent variable related to hypothesis 3, the announcement returns is adjusted by shortening the event window to seven and three days, presented in Appendix 4. I additionally compute the outcome variable CAR through the mean-adjusted model instead of the market model using an eleven-, seven- and three-day event window, shown in Appendix 5. Both alternative measurements yield likewise results to Table 5, providing additional insights but again only with some statistical evidence for the positive effect of policy EPU in the acquiror nation.

4.4.4 - Regulated Industries

As aforementioned, the presence of regulated industries may overshadow the effects of EPU on deal outcomes, given their higher exposure to policy risks. Therefore, Appendix 6 presents a sub-analysis, excluding the most heavily regulated industries, to assess the impact of EPU on deal outcomes beyond the influence of the highly regulated sectors.

Consistently with previous findings, Table 8 reveals a negative (positive) effect of EPU in the target's (acquiror's) nation on deal premium as well as announcement returns, with similar levels of significance. Notably, all coefficients exhibit a slight decrease in value, indicating a relatively milder reaction of non-regulated industries to EPU as anticipated. I conclude that even after excluding regulated industries, the results demonstrate consistent findings, where the coefficients associated with EPU maintain the same direction, but only in slightly reduced magnitudes.

5 - Conclusion and Implications

In this study, I attempted to gain insights into the impact of EPU on the cross-border M&A activity of 18 sample countries over the period 2004 to 2022. I examine the effect of EPU on (1) inbound and outbound M&A deal flow, (2) deal premia and (3) announcement returns to answer the following research question:

RQ: What is the effect of Economic Policy Uncertainty (EPU) on cross-border M&A activity and performance?

Prior literature studying the effect of EPU on cross-border M&A has shown to be mixed. EPU may have an impeding effect on M&A activity since firms tend to delay or avoid especially irreversible investments and rely on more conservative strategies (Al-Thaqeb & Algharabali, 2019; Holburn & Zelner, 2010). At the transaction level, elevated levels of EPU in a country lead to deviations in bargaining power and altering of investment choices which affect performance levels (Lee, 2018; Nguyen & Phan, 2017). On the other hand, EPU may be used as a risk management strategy, where firms are encouraged to acquire targets with lower levels of EPU as a risk-reducing strategy (Cao et al., 2019; Le & Zak, 2006). Given that EPU affects the financial macroconditions of a country, but may also complicate synergy expectations (Gorbenko & Malenko, 2014; Hege et al., 2009), strategic and financial acquirors may have different attitudes towards EPU. I contribute to the existing literature by studying the effect of EPU on cross-border M&A, where I provide a deeper understanding of the discrepancies in sensitivity of different types of acquirors to EPU in terms of performance and bidding behaviour. This research has important implications for financial and strategic shareholders who may alter their strategy in premium offers and target choices when associated with EPU. Moreover, policymakers and professionals can anticipate uncertain environments and build strategies to cope with the macroeconomic consequences of EPU.

I employ the index created by Baker et al. (2016) to measure EPU and obtain deal information from the Refinitiv Eikon database. I retrieve additional data from Datastream, World Bank, World Factbook, CEPPI, La Porta et al. (2008) and Kenneth French (2023) databases. After performing panel regressions on country-level data, I find significant evidence that EPU in target and acquiror nation reduces M&A activity. Furthermore, on a transaction level, I find contrasting effects of EPU in the target and acquiror nation on deal premium offer, which is consistent with extant literature on relative bargaining power to compensate for country risk through favourable deal terms. Here,

financial acquirors tend to exhibit greater sensitivity to EPU in decreasing their offer premia. This may be driven by their short-term vision and anticipation of the rising cost of capital. Finally, the acquiror announcement returns, measured by their eleven-day CAR, are negatively (positively) related to the EPU of the target (acquiror) nation. These findings are in line with the theoretical argument that target EPU hampers investment growth, while acquiror EPU increases discretion in choices of M&A deals resulting in better expected outcomes. Strategic acquirors' performance in my sample is impacted more negatively by EPU, which may be attributable to their reliance on future synergy potential rather than short-term returns. However, as the deal analyses are not statistically significant, I find insufficient evidence to support my findings and prior literature. The results are robust to alternative variable measurements, instrumental variable analysis, and the exclusion of highly regulated industries.

5.1 - Shortcomings and Limitations

This research is subject to several limitations. First, the sample size is constrained by a variety of factors. The availability of EPU indices was limited to 28 countries, with individual variations in time periods. Additionally, the sample size is substantially reduced as unlisted firms do not report shareholder returns and deal premia offers. The exclusion of unlisted firms also resulted in a substantial reduction in sample size of financial acquirors, as those mainly engage in private transactions. A similar study by Paudyal et al. (2021) encompasses monthly rather than yearly observations, which may yield more significant results given the possibility of higher variability in data. Contrarily, this may also lead to more noise in the result estimates. Thus, I advise future research to explore alternative data sources to gather additional information on private M&A deals.

A second limitation is that, despite the widespread recognition of the BBD index as comprehensive measure of EPU, the index has been criticized. For instance, the measurement method is prone to omitting important signals of risk due to the presence of sarcasm, metaphors, and ambiguity in newspapers, leading to misinterpretation of the text and misspecification of the index rates. Furthermore, the index may capture other non-policy related economic uncertainty, such as stock market volatility or currency uncertainty that may cause a potential error-in-measurement problem that biases the results. As a solution, I suggest performing additional analyses of the stock market or currency volatility index as explanatory variables to test for the robustness of the results.

A limitation regarding my methodology is the complexity of differentiating between the motives of strategic and financial acquirors. The financial acquiror flag serves as an indicator to distinguish acquisitions conducted by financial acquirors with financial motives but fails to encompass the nuanced landscape of strategic acquirors who may also bear underlying financial motives. Therefore, further work is needed to gain deeper insights into the implications of these groups. Furthermore, financial and strategic acquirors could be classified through dominant strategies such as industry diversification or correlating cash flows, rather than solely relying on their motives.

Even though this study includes the most prominent control variables from prior literature, the findings may still be prone to omitted variable bias. As a solution, one could include additional variables that may influence the M&A environment as a country's business environment and stock market capitalization, such as in the studies of [Gregoriou et al. \(2021\)](#) and [Paudyal et al. \(2021\)](#). Similarly, my cross-sectional research on deal premia and announcement returns is susceptible to self-selection bias, given that M&As are managers' choices rather than random selections. The Heckman two-step self-selection model has proven to mitigate this concern (Nguyen & Phan, 2017; Paudyal et al., 2021). However, this test requires the application of a probit model, which was not employed in this study. Hence, future endeavours focusing on investigating factors such as the likelihood of engaging in acquisitions could employ the Heckman model to correct for self-selection bias.

Finally, further recommendations for academics that are interested in investigating the impact of EPU on cross-border M&A activity. Scholars could expand upon the metrics used in this study. For instance, while this research focuses on short-term acquiror shareholder value, future endeavours could explore long-term value creation through the buy-and-hold strategy as employed by [Bonaime et al. \(2018\)](#). Additionally, investigating other metrics such as the chances of stock or cash payment, as well as the likelihood of engaging in M&A, would contribute to a more comprehensive understanding of the cross-border M&A landscape under elevated EPU levels.

Lastly, future research could delve into exploring the empirical implications of the economic mechanisms proposed in existing literature, such as hedging motives or investment irreversibility, that influence the decision-making process surrounding cross-border M&A activity during high EPU. Such endeavours would significantly contribute to countries' ability to better anticipate policy uncertainties and comprehend the M&A landscape in a more profound manner.

Appendices

Appendix 1: Deals by Acquiror and Target per Country

Country	Deals by Acquiror Nation			Deals by Target Nation		
	Domestic	Cross-Border	Total	Domestic	Cross-Border	Total
Australia (AU)	6,340	977	7,317	6,340	1,042	7,382
Brazil (BR)	1,761	74	1,835	1,761	373	2,134
Canada (CA)	6,022	2,847	8,869	6,022	1,500	7,522
China (CN)	14,839	812	15,651	14,839	1,462	16,301
Colombia (CO)	115	27	142	115	128	243
France (FR)	1,469	753	2,222	1,469	838	2,307
Germany (DE)	1,069	661	1,730	1,069	1,101	2,170
Greece (GR)	160	16	176	160	46	206
Hong Kong (HK)	2,239	1,437	3,676	2,239	521	2,760
India (IN)	1,457	380	1,837	1,457	271	1,728
Japan (JP)	6,154	678	6,832	6,154	223	6,377
Mexico (MX)	329	95	424	329	305	634
Netherlands (NL)	364	509	873	364	576	940
Russia (RU)	1,195	61	1,256	1,195	115	1,310
South Korea (KR)	3,860	317	4,177	3,860	224	4,084
Spain (ES)	1,365	355	1,720	1,365	673	2,038
United Kingdom (GB)	8,302	2,640	10,942	8,302	2,257	10,559
United States (US)	31,966	4,257	36,223	31,966	5,241	37,207
Total	89,006	16,896	105,902	89,006	16,896	105,902

Appendix 2: Variables, Definitions, and Sources

APPENDIX 2 Variables Definitions and Measurements		
Name	Definition/Measurement	Source
<i>Dependent Variables</i>		
Inbound CBA	The logarithm of the aggregate number of completed M&A deals for each target country per month	Refinitiv Eikon
Outbound CBA	The logarithm of the aggregate number of completed M&A deals per acquiror country	Refinitiv Eikon
4-week Deal Premium	Percentage difference between the offer price and the target's stock price 4 weeks before the announcement	Refinitiv Eikon
Acquiror market model CAR [-5,+5]	Acquiror's cumulative abnormal return following market model expected returns over an 11-day event window. MSCI World Index serves as the market index.	Datastream
<i>Explanatory Variables</i>		
Ln(EPU) _{tgt}	The news-based BBD index with an original monthly frequency of the target nation. I use the natural logarithm of the 12 arithmetic mean over 12 months to obtain yearly values.	Baker et al. (2016)
Ln(EPU) _{acq}	The news-based BBD index with an original monthly frequency of the acquiror nation. I use the natural logarithm of the 12 arithmetic mean over 12 months to obtain yearly values.	Baker et al. (2016)
EPU_M	The 1 year lagged median of the yearly BBD index across 18 sample countries.	Baker et al. (2016)
EPU low	Dummy variable equal to 1 if the EPU of a country is below the EPU median.	Baker et al. (2016)
EPU high	Dummy variable equal to 1 if the EPU of a country is above the EPU median.	Baker et al. (2016)

<i>Moderating Variable</i>		
Financial Acquiror	Dummy variable equal to one if the acquiror is financial (primary industry SIC code starting with 6), zero if the acquiror is strategic	Refinitiv Eikon
<i>Country-characteristic Control Variables</i>		
Real GDP growth	The annual real gross domestic product (GDP) growth rate per capita (in USD).	World Bank
Ln (GDP/Capita)	The natural logarithm of the annual GDP at constant USD prices (2015) per capita.	World Bank
Trade Openness	The annual sum of the annual trade (import-export) of goods and services scaled over the GDP.	World Bank
Inflation	The annual consumer price index of a country.	World Bank
Exchange rate	The natural logarithm of the annual average exchange rate of local currency units relative to USD.	World Bank
Quality of Institutions	Consists of 3 dimensions, Corruption, Government Effectiveness and Political Stability, to measure a country's political environment and government quality. Each dimension gives a country's score as the logarithm of the aggregate indicator, ranging from -2.5 to 2.5.	World Bank
Regulatory Quality	Ability of the government to formulate and implement policies, giving a country's score on the logarithm of the aggregate indicator, ranging from -2.5 to 2.5.	World Bank
<i>Cross-border Control Variables</i>		
Ln (Geographic Distance)	The natural log of geographical distance in kilometres is calculated following the great circle formula, using the latitudes and longitudes of the most populated cities of both countries.	CEPII
Border	Dummy variable equal to one if target and acquiror countries are contiguous, zero otherwise.	CEPII
Legal Origin	Dummy variable equal to one if target and acquiror countries have a similar legal origin. Legal origin is classified in common or civil law, where the latter is further split into French, German or Scandinavian.	La Porta et al. (2008)
Language	Dummy variable equal to one if target and acquiror speak the same official language, zero otherwise.	World Factbook
Religion	Dummy variable equal to one if target and acquiror have the same primary religion.	World Factbook
<i>Acquiror Firm-characteristic Control Variables</i>		
Firm Size	The natural logarithm of the book value of total assets in USD (WC02999).	Datastream
Cash Holdings	Ratio of cash and equivalents (WC02001) over the total value of assets (WC02999).	Datastream
Leverage	Ratio of the sum of debt in current liabilities (WC03051) and long-term debt (WC03251) to total value of assets of the last twelve months.	Datastream
Market-to-Book	The market value of common equity over the book value of equity (MTBV).	Datastream
<i>Deal-characteristic Control Variables</i>		
Deal Size	The natural logarithm of the total value of consideration paid by the acquiror in USD, excluding fees and expenses.	Refinitiv Eikon
Cash	Dummy variable equal to one if an M&A deal is fully funded by cash, zero otherwise.	Refinitiv Eikon
Stock	Dummy variable equal to one if an M&A deal is fully funded by stock, zero otherwise.	Refinitiv Eikon
Diversifying Deal	Dummy variable equal to one if target and acquirer do not have similar Fama-French 49 industry groups, zero otherwise.	Kenneth R. French (2023)
Competing Offer	Dummy variable equal to one if there are reported competing bids for the same deal, zero otherwise.	Refinitiv Eikon
Tender Offer	Dummy variable equal to one if the deal is defined as a tender offer, zero otherwise.	Refinitiv Eikon

Appendix 3: Alternative Measures of Deal Premium

APPENDIX 3				
Robustness Deal Premium				
This Table presents the effect of EPU on alternative measures of Deal Premia. The dependent variable is measured as the percentage difference of the offer price over the target's stock price, for columns (1) and (2) 1 week prior and for columns (3) and (4) 1 day prior to the deal announcement. EPU is measured for the target country in columns (1) and (3), with target country-fixed effects, and measured for the acquiror country in columns (2) and (4), using acquiror country-fixed effects. Year-, Country and Industry-fixed effects are included. Robust standard errors adjusted for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.				
VARIABLES	Premium 1 Week Prior		Premium 1 day Prior	
	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-1.027 (6.197)		-3.785 (5.192)	
Ln(EPU) _{acq}		0.422 (9.061)		0.522 (9.106)
Constant	-5.839 (64.378)	272.114** (123.266)	-10.826 (53.928)	290.097** (122.851)
Control Variables	Yes	Yes	Yes	Yes
Country, Year & Industry FE	Yes	Yes	Yes	Yes
Observations	669	669	672	672
Adjusted R-squared	0.071	0.065	0.076	0.081

Appendix 4: Alternative Event Windows Market Model Cumulative Abnormal Returns (CARs)

APPENDIX 4				
Robustness Event Windows Market Model CARs				
This Table presents the effect of Economic Policy Uncertainty on the announcement returns of the acquiror using alternative event windows. The dependent variable is the cumulative abnormal return (CAR), estimated over a 7-day event window for columns (1) and (2), and 3-day event window in columns (3) and (4). The main explanatory variable is measured for the target country in columns (1) and (3), with target country-fixed effects, and measured for the acquiror country in columns (2) and (4), using acquiror country-fixed effects. The results include year-, country-, and industry-fixed effects. Robust standard errors are included to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.				
VARIABLES	CAR[-3,+3]		CAR[-1,+1]	
	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-0.009 (0.021)		-0.010 (0.015)	
Ln(EPU) _{acq}		0.028 (0.020)		0.025* (0.013)
Constant	-1.226 (1.450)	-3.058 (2.889)	1.676 (2.129)	-1.329 (1.131)
Control Variables	Yes	Yes	Yes	Yes
Country, Year & Industry FE	Yes	Yes	Yes	Yes
Observations	786	786	786	786
Adjusted R-squared	-0.022	-0.011	0.038	0.050

Appendix 5: Mean Adjusted Model Cumulative Abnormal Returns (CARs)

APPENDIX 5 Mean Adjusted CARs						
This Table presents the effect of Economic Policy Uncertainty on the announcement returns of the acquiror using the mean adjusted model and varying event windows. The dependent variable is the cumulative abnormal return (CAR), estimated over an 11-day event window for columns (1) and (2), and 7-day event window in columns (3) and (4), and 3-day event window in columns (5) and (6). The main explanatory variable is measured for the target country in columns (1) and (3), with target country-fixed effects, and measured for the acquiror country in columns (2) and (4), using acquiror country-fixed effects. The results include year-, country-, and industry-fixed effects. Robust standard errors are included to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.						
VARIABLES	CAR[-5,+5]		CAR[-3,+3]		CAR[-1,+1]	
	(1)	(2)	(3)	(4)	(5)	(6)
Ln(EPU) _{tgt}	-0.020 (0.024)		-0.013 (0.019)		-0.009 (0.014)	
Ln(EPU) _{acq}		0.047** (0.024)		0.020 (0.020)		0.022 (0.014)
Constant	4.078 -3.728	1.122 -2.198	4.04 -2.687	-0.519 -1.631	2.351 -1.942	-1.964* -1.149
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Country, Year &	Yes	Yes	Yes	Yes	Yes	Yes
Observations	786	786	786	786	786	786
Adjusted R-squared	-0.019	-0.017	0.015	-0.005	0.045	0.033

Appendix 6: Robustness Excluding Highly Regulated Industries

APPENDIX 6 Non-Regulated Industries – Deal Premium and CARs Analysis				
This table examines the effect of EPU on the offer premium, using a subsample excluding heavily regulated industries (SIC codes 4900-4999, 6000-6999). The dependent variable is measured as the percentage difference of the offer price over a target's stock price four weeks prior to the deal announcement. Country, year, and industry fixed effects are included. Robust standard errors in parentheses to adjust for heteroskedasticity. *** p<0.01, ** p<0.05, * p<0.1.				
VARIABLES	Deal Premium (4 weeks)		CAR [-5,+5]	
	(1)	(2)	(3)	(4)
Ln(EPU) _{tgt}	-0.0180 (9.803)		-0.006 (0.027)	
Ln(EPU) _{acq}		0.304 (9.991)		0.049* (0.025)
Constant	-5.335 (82.278)	315.515** (138.875)	0.984 (4.085)	0.939 (2.148)
Control Variables	Yes	Yes	Yes	Yes
Country, Industry and Year FE	Yes	Yes	Yes	Yes
Observations	631	631	732	732
Adjusted R-squared	0.069	0.062	-0.017	-0.009

References

- Acemoglu, D., Johnson, S., Robinson, J., & Thaicharoen, Y. (2003). Institutional causes, macroeconomic symptoms: Volatility, crises and growth. *J. Monet. Econ.*, 50(1), 49–123. [https://doi.org/10.1016/s0304-3932\(02\)00208-8](https://doi.org/10.1016/s0304-3932(02)00208-8)
- Ackert, L. F., & Athanassakos, G. (1997). Prior Uncertainty, Analyst Bias, and Subsequent Abnormal Returns. *Journal of Financial Research*, 20(2), 263–273. <https://doi.org/10.1111/j.1475-6803.1997.tb00248.x>
- Ahern, K. R., Daminelli, D., & Fracassi, C. (2015). Lost in translation? The effect of cultural values on mergers around the world. *J. Financ. Econ.*, 117(1), 165–189. <https://doi.org/10.1016/j.jfineco.2012.08.006>
- Akey, P., & Lewellen, S. (2016). Policy uncertainty, political capital, and firm risk-taking. *SSRN Electron. J.* <https://doi.org/10.1016/j.jeca.2019.e00133>
- Al-Thaqeb, S. A., & Algharabali, B. G. (2019). Economic policy uncertainty: A literature review. *J. Econ. Asymmetries*, 20(e00133), e00133. <https://doi.org/10.1016/j.jeca.2019.e00133>
- Azqueta-Gavaldon, A., Hirschbühl, D., Onorante, L., & Saiz, L. (2020). Economic policy uncertainty in the euro area: An unsupervised machine learning approach. *SSRN Electron. J.* <https://doi.org/10.2139/ssrn.3516756>
- Baker, S. R., Bloom, N., & Davis, S. J. (2016). Measuring economic policy uncertainty. *Q. J. Econ.*, 131(4), 1593–1636. <https://doi.org/10.1093/qje/qjw024>
- Bargeron, L., Schlingemann, F., Stulz, R., & Zutter, C. (2008). Why do private acquirers pay so little compared to public acquirers? ☆. *J. Financ. Econ.*, 89(3), 375–390. <https://doi.org/10.1016/j.jfineco.2007.11.005>

- Belke, A., Dubova, I., & Osowski, T. (2018). Policy uncertainty and international financial markets: The case of Brexit. *Appl. Econ.*, 50(34–35), 3752–3770. <https://doi.org/10.1080/00036846.2018.1436152>
- Bernanke, B. S. (1983). Irreversibility, uncertainty, and cyclical investment. *Q. J. Econ.*, 98(1), 85. <https://doi.org/10.2307/1885568>
- Bhagwat, V., Dam, R., & Harford, J. (2016). The real effects of uncertainty on merger activity. *Rev. Financ. Stud.*, 29(11), 3000–3034. <https://doi.org/10.1093/rfs/hhw061>
- Bloom, N. (2009). The Impact of Uncertainty Shocks. *Econometrica*, 77(3), 623–685. <https://doi.org/10.3982/ECTA6248>
- Bloom, N., Bond, S., & Van Reenen, J. (2007). Uncertainty and Investment Dynamics. *The Review of Economic Studies*, 74(2), 391–415. <https://doi.org/10.1111/j.1467-937X.2007.00426.x>
- Bonaime, A., Gulen, H., & Ion, M. (2018). Does policy uncertainty affect mergers and acquisitions? *J. Financ. Econ.*, 129(3), 531–558. <https://doi.org/10.1016/j.jfineco.2018.05.007>
- Bordo, M. D., Duca, J. V., & Koch, C. (2016). Economic policy uncertainty and the credit channel: Aggregate and bank-level U.S. evidence over several decades. *J. Fin. Stab.*, 26, 90–106. <https://doi.org/10.1016/j.jfs.2016.07.002>
- Brogaard, J., & Detzel, A. (2015). The Asset-Pricing Implications of Government Economic Policy Uncertainty. *Management Science*, 61(1), 3–18. <https://doi.org/10.1287/mnsc.2014.2044>
- Cao, C., Li, X., & Liu, G. (2019). Political uncertainty and cross-border acquisitions. *Rev. Financ.*, 23(2), 439–470. <https://doi.org/10.1093/rof/rfx055>
- Castro-Iragorri, C. (2019). Does the market model provide a good counterfactual for event studies in finance? *Fin. Mark. Portf. Manag.*, 33(1), 71–91.

- Chiarella, C., & Ostinelli, D. (2020). Financial or strategic buyers: Who is at the gate? *International Review of Economics & Finance*, 67, 393–407. <https://doi.org/10.1016/j.iref.2020.02.005>
- Choi, S.-W. (2009). The Effect of Outliers on Regression Analysis: Regime Type and Foreign Direct Investment. *Quarterly Journal of Political Science*, 4(2), 153–165. <https://doi.org/10.1561/100.00008021>
- Čirjevskis, A. (2019). The role of dynamic capabilities as drivers of business model innovation in mergers and acquisitions of technology-advanced firms. *J. Open Innov.*, 5(1), 12. <https://doi.org/10.3390/joitmc5010012>
- Davis, S. (2016). *An index of global economic policy uncertainty*. National Bureau of Economic Research.
- Degiannakis, S., & Filis, G. (2019). Forecasting European economic policy uncertainty. *Scott. J. Polit. Econ.*, 66(1), 94–114.
- Denis, D. J., Denis, D. K., & Yost, K. (2002). Global diversification, industrial diversification, and firm value. *J. Finance*, 57(5), 1951–1979. <https://doi.org/10.1111/0022-1082.00485>
- Dittmar, A., Li, D., & Nain, A. (2012). It pays to follow the leader: Acquiring targets picked by private equity. *J. Fin. Quant. Anal.*, 47(5), 901–931.
- Duchin, R., & Schmidt, B. (2013). Riding the merger wave: Uncertainty, reduced monitoring, and bad acquisitions. *J. Financ. Econ.*, 107(1), 69–88. <https://doi.org/10.1016/j.jfineco.2012.07.003>
- Erel, I., Liao, R. C., & Weisbach, M. S. (2012). Determinants of cross-border mergers and acquisitions. *J. Finance*, 67(3), 1045–1082. <https://doi.org/10.1111/j.1540-6261.2012.01741.x>

- Farooq, O., & Ahmed, N. (2019). Dividend policy and political uncertainty: Evidence from the US presidential elections. *Res. Int. Bus. Fin.*, 48, 201–209. <https://doi.org/10.1016/j.ribaf.2019.01.003>
- Ferreira, M. A., Massa, M., & Matos, P. (2010). Shareholders at the Gate? Institutional Investors and Cross-Border Mergers and Acquisitions. *Review of Financial Studies*, 23(2), 601–644. <https://doi.org/10.1093/rfs/hhp070>
- Fidrmuc, J. P., Roosenboom, P., Paap, R., & Teunissen, T. (2012). One size does not fit all: Selling firms to private equity versus strategic acquirers. *J. Corp. Fin.*, 18(4), 828–848. <https://doi.org/10.1016/j.jcorpfin.2012.06.006>
- García-Canal, E., & Guillén, M. F. (2008). Risk and the strategy of foreign location choice in regulated industries. *Strategic Management Journal*, 29(10), 1097–1115. <https://doi.org/10.1002/smj.692>
- Gemson, J., Gautami, K. V., & Thillai Rajan, A. (2012). Impact of private equity investments in infrastructure projects. *Util. Policy*, 21, 59–65. <https://doi.org/10.1016/j.jup.2011.12.001>
- Gorbenko, A. S., & Malenko, A. (2014). Strategic and financial bidders in takeover auctions. *J. Finance*, 69(6), 2513–2555. <https://doi.org/10.1111/jofi.12194>
- Gregoriou, A., Nguyen, B. D., Nguyen, T. D., Le, H., & Hudson, R. (2021). Economic policy uncertainty and cross-border mergers and acquisitions. *International Review of Financial Analysis*, 78, 101911. <https://doi.org/10.1016/j.irfa.2021.101911>
- Groh, A. P. (2009). PRIVATE EQUITY IN EMERGING MARKETS. *PRIVATE EQUITY IN EMERGING MARKETS*.
- Gulen, H., & Ion, M. (2016). Policy Uncertainty and Corporate Investment. *The Review of Financial Studies*, 29(3), 523–564. <https://doi.org/10.1093/rfs/hhv050>

- Hankins, K. W. (2006). *ARE ACQUISITIONS AN OPERATIONAL HEDGE? THE INTERACTION OF FINANCIAL AND OPERATIONAL HEDGING*.
- Healy, P. M., Palepu, K. G., & Ruback, R. S. (1997). Which takeovers are profitable? Strategic or financial? *Sloan Manage. Rev.*, 38(4), 45–57.
- Hege, U., Lovo, S., Slovin, M. B., & Sushka, M. E. (2009). Asset Sales and the Role of Buyers: Strategic Buyers versus Private Equity. *SSRN Electronic Journal*.
<https://doi.org/10.2139/ssrn.1787465>
- Hege, U., Lovo, S., Slovin, M. B., & Sushka, M. E. (2012). Asset sales and the role of buyers: Strategic buyers versus private equity. *SSRN Electron. J.*
- Henisz, W. J., & Zelner, B. A. (2001). The Institutional Environment for Telecommunications Investment. *Journal of Economics & Management Strategy*, 10(1), 123–147.
<https://doi.org/10.1111/j.1430-9134.2001.00123.x>
- Holburn, G. L. F., & Zelner, B. A. (2010). Political capabilities, policy risk, and international investment strategy: Evidence from the global electric power generation industry. *Strategic Manage. J.*, 31(12), 1290–1315. <https://doi.org/10.1002/smj.860>
- Hussain, T., Tunyi, A. A., Sufyan, M., & Shahab, Y. (2022). Powerful bidders and value creation in M&As. *International Review of Financial Analysis*, 81, 102076.
<https://doi.org/10.1016/j.irfa.2022.102076>
- Hyun, H.-J., & Kim, H. H. (2010). The determinants of cross-border M&As: The role of institutions and financial development in the gravity model. *World Econ.*, 33(2), 292–310.
<https://doi.org/10.1111/j.1467-9701.2009.01224.x>
- Julio, B., & Yook, Y. (2012). Political uncertainty and corporate investment cycles. *J. Finance*, 67(1), 45–83. <https://doi.org/10.1111/j.1540-6261.2011.01707.x>

- Kang, N.-H., & Johansson, S. (2000). *Cross-border mergers and acquisitions* (OECD Science, Technology and Industry Working Papers). Organisation for Economic Co-Operation and Development (OECD). <https://doi.org/10.1787/137157251088>
- Keown, A. J., & Pinkerton, J. M. (1981). Merger announcements and insider trading activity: An empirical investigation. *J. Finance*, 36(4), 855–869.
- Klomp, J., & de Haan, J. (2009). Political institutions and economic volatility. *Eur. J. Polit. Econ.*, 25(3), 311–326.
- Kozol, E., & Oran, J. (2021). FINANCIAL RISK MANAGEMENT AND FIRM PERFORMANCE: EVIDENCE FROM EUROPEAN CROSS-BORDER MERGERS AND ACQUISITIONS. *Öneri Dergisi*, 16(56), 701–718. <https://doi.org/10.14783/maruoneri.788480>
- Kundu, S., & Paul, A. (2022, July). *Effect of economic policy uncertainty on stock market return and volatility under heterogeneous market characteristics—ScienceDirect*. <https://www.sciencedirect.com/science/article/pii/S1059056022000685>
- La Porta, R., Lopez-de-Silanes, F., & Shleifer, A. (2008). The economic consequences of legal origins. *J. Econ. Lit.*, 46(2), 285–332. <https://doi.org/DOI:10.1257/jel.46.2.285>
- La Porta, R., Lopez-de-Silanes, F., Shleifer, A., & Vishny, R. W. (2001). Law and Finance. In *Corporate Governance* (pp. 26–68). Springer Berlin Heidelberg. https://link.springer.com/chapter/10.1007/978-3-642-59499-1_3
- Le, Q. V., & Zak, P. J. (2006). Political risk and capital flight. *J. Int. Money Finance*, 25(2), 308–329. <https://doi.org/10.1016/j.jimonfin.2005.11.001>
- Lee, K. H. (2018). Cross-border mergers and acquisitions amid political uncertainty: A bargaining perspective. *Strategic Management Journal*, 39(11), 2992–3005. <https://doi.org/10.1002/smj.2944>

- Makaew, T. (2012). *Waves of International Mergers and Acquisitions* (SSRN Scholarly Paper No. 1786989). <https://doi.org/10.2139/ssrn.1786989>
- McBeath, I., & Bacha, J. (2001). Mergers and acquisitions: A consideration of the drivers and hurdles. *J. Commer. Biotechnol.*, 8(2). <https://doi.org/10.5912/jcb425>
- McDonald, R., & Siegel, D. (1986). The value of waiting to invest. *Q. J. Econ.*, 101(4), 707. <https://doi.org/10.2307/1884175>
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *J. Finance*, 29(2), 449–470. <https://doi.org/10.1111/j.1540-6261.1974.tb03058.x>
- Myers, S. C., & Majluf, N. S. (1984). Corporate financing and investment decisions when firms have information that investors do not have. *J. Financ. Econ.*, 13(2), 187–221. [https://doi.org/10.1016/0304-405x\(84\)90023-0](https://doi.org/10.1016/0304-405x(84)90023-0)
- Nguyen, N. H., & Phan, H. V. (2017). Policy Uncertainty and Mergers and Acquisitions. *Journal of Financial and Quantitative Analysis*, 52(2), 613–644. <https://doi.org/10.1017/S0022109017000175>
- Pástor, L., & Veronesi, P. (2013). Political uncertainty and risk premia. *J. Financ. Econ.*, 110(3), 520–545. <https://doi.org/10.1016/j.jfineco.2013.08.007>
- Paudyal, K., Thapa, C., Koirala, S., & Aldhawyan, S. (2021). Economic policy uncertainty and cross-border mergers and acquisitions. *Journal of Financial Stability*, 56, 100926. <https://doi.org/10.1016/j.jfs.2021.100926>
- Phan, H. V., Nguyen, N. H., Nguyen, H. T., & Hegde, S. (2019). Policy uncertainty and firm cash holdings. *J. Bus. Res.*, 95, 71–82. <https://doi.org/10.1016/j.jbusres.2018.10.001>
- Rodgers, D. E. (2015). Strategic or financial: Which buyer pays more? In *FOCUS Investment Banking LLC*. <https://focusbankers.com/strategic-or-financial-which-buyer-pays-more/>

- Rossi, S., & Volpin, P. F. (2004). Cross-country determinants of mergers and acquisitions. *Journal of Financial Economics*, 74(2), 277–304. <https://doi.org/10.1016/j.jfineco.2003.10.001>
- Serdar Dinc, I., & Erel, I. (2013). Economic nationalism in mergers and acquisitions. *J. Finance*, 68(6), 2471–2514. <https://doi.org/10.1111/jofi.12086>
- Sha, Y., Kang, C., & Wang, Z. (2020). Economic policy uncertainty and mergers and acquisitions: Evidence from China. *Economic Modelling*, 89, 590–600. <https://doi.org/10.1016/j.econmod.2020.03.029>
- Shams, S., Gunasekarage, A., & Velayutham, E. (2022). Economic policy uncertainty and acquisition performance: Australian evidence. *International Review of Economics & Finance*, 78, 286–308. <https://doi.org/10.1016/j.iref.2021.12.004>
- Tran, Q. T. (2021). Economic policy uncertainty and cost of debt financing: International evidence. *The North American Journal of Economics and Finance*, 57, 101419. <https://doi.org/10.1016/j.najef.2021.101419>
- Uddin, M., & Boateng, A. (2011). Explaining the trends in the UK cross-border mergers & acquisitions: An analysis of macro-economic factors. *International Business Review*, 20(5), 547–556. <https://doi.org/10.1016/j.ibusrev.2010.11.003>
- Xie, E., Reddy, K. S., & Liang, J. (2017). Country-specific determinants of cross-border mergers and acquisitions: A comprehensive review and future research directions. *J. World Bus.*, 52(2), 127–183. <https://doi.org/10.1016/j.jwb.2016.12.005>
- Youssef, M., Mokni, K., & Ajmi, A. N. (2021). Dynamic connectedness between stock markets in the presence of the COVID-19 pandemic: Does economic policy uncertainty matter? *Financ. Innov.*, 7(1), 13. <https://doi.org/10.1186/s40854-021-00227-3>

Zhang, G., Han, J., Pan, Z., & Huang, H. (2015). Economic policy uncertainty and capital structure choice: Evidence from China. *Econ. Syst.*, 39(3), 439–457.
<https://doi.org/10.1016/j.ecosys.2015.06.003>